

A Project Report on **Water Pollution Monitoring Using RC Boat**

Submitted by

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Submitted in partial fulfilment of the Requirements for the award of the Degree of

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)

SEMESTER V EXAMINATION



DEPARTMENT OF INFORMATION TECHNOLOGY
THAKUR COLLEGE OF SCIENCE AND COMMERCE
(Permanently Affiliated to University of Mumbai)
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CERTIFICATE

This is to certify that the project titled “**Water Pollution Monitoring Using RC Boat**”. Undertaken at the “Thakur College of Science and Commerce” by **Mr. SRIKRITH SHRIYAN** Seat no. **25142** is submitted in partial fulfilment of the requirements for the award of degree of BACHELOR OF SCIENCE in INFORMATION TECHNOLOGY SEM V Examination and does not form part of any other course undergone by the candidate.

It is further certified that we have completed all required phases of the project.

Project Guide

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CERTIFICATE

This is to certify that the project titled “Water Pollution Monitoring Using RC Boat”. Undertaken at the “Thakur College of Science and Commerce” by Mr Swapnil Gaur Seat no. 25275 is submitted in partial fulfilment of the requirements for the award of degree of BACHELOR OF SCIENCE in INFORMATION TECHNOLOGY SEM V Examination and does not form part of any other course undergone by the candidate.

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Project Guide

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Certificate of Authenticated Work



Project Information

B.Sc.

Water Pollution Monitoring RC Boat



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Description Technology Modules Info

Modules

No modules yet!

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Abstract

Water pollution is currently one of the most urgent environmental issues in the modern world, influencing aquatic ecosystems, human health, and general ecosystem balance. Conventional monitoring of water quality is usually through manual sampling and analysis in the laboratory, which are labor-intensive, expensive, and unsafe in the case of contaminated or hard-to-reach water bodies. To overcome these constraints, the project proposes the design and implementation of an affordable and effective water pollution monitoring system utilizing a remotely controlled (RC) boat. The system is developed based on an Arduino Uno microcontroller as the processing unit, which includes a turbidity sensor for measuring the water quality, two DC motors with fans for movement, and an HC-05 Bluetooth module for wireless communication and control. RC boat is made with light materials like thermocol and cardboard, so it is portable, lightweight, and inexpensive.

The turbidity sensor monitors the water clarity, which is a key parameter to show suspended particles and contaminants. The data measured is processed with the help of Arduino and sent to a mobile phone through Bluetooth for real-time monitoring. Also, the same Bluetooth module enables the user to maneuver the movement of the boat in forward and backward direction, thus able to travel to various reaches of a body of water. The dual mode of monitoring and mobility provides flexible and effective data collection over larger areas than with manual sampling.

This project showcases an innovative but pragmatic solution to water pollution monitoring by integrating sensor technology and remote-controlled mobility. It reduces human exposure to risk, lowers monitoring effort, and offers real-time information for water quality evaluation. The system holds enormous scope for application in lakes, rivers, and reservoirs and can even be used as a platform for additional development like integrating multiple sensors, GPS-based tracking, or cloud-based storage. As a whole, the pollution monitoring RC boat is part of sustainable environmental management as well as encouraging the adoption of smart, technology-based solutions to everyday issues.

Acknowledgment

The success and final outcome of this project required a lot of guidance and assistance from many people and I am extremely fortunate to have got this all along the completion of my project work. Whatever I have done is only due to such guidance and assistance and I would not forget to thank them.

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Srikrith Shriyan

Rishab Desai

Swapnil Gaur

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1. INTRODUCTION

Pollution of water is one of the most important environmental issues that are encountered in today's world. Rivers, lakes, and ponds are exposed to toxic waste and pollutants because of industrialization, urbanization, and human activities. Suspended solids, pathogens, and chemical residues are present in polluted water, which harms aquatic life and leads to serious health risks for humans. Contaminated water can lead to diseases, interfere with biodiversity, and lower the quality of life. Conventional water quality monitoring has generally been based on manual sampling by hand and subsequent laboratory analysis. Although efficient in closed environments, such methods are time-consuming, expensive, and not equipped to generate real-time data. In addition, accessing particular areas of water bodies like contaminated areas, industrial outlet locations, or deeper areas becomes dangerous for crew members, making traditional methods unfeasible for massive or real-time monitoring. Such constraints place an imperative on finding new, safe, and economical ways to measure levels of pollution in water bodies. In order to address these issues, the idea of a remotely operated boat equipped with a turbidity sensor has been established. The RC boat is used as a transportable platform with the ability to travel over various areas of a water body and continuously monitoring the water clarity. One of the most significant parameters of pollution is turbidity, as it indicates the concentration of suspended particles in the water, which commonly consist of sediments, organic matter, and pollutants. A larger turbidity reading is a direct indication of more contamination, so it is an effective measure of water quality in the field. The turbidity sensor's readings are gathered by a microcontroller on the boat, which processes the data and sends it wirelessly via a Bluetooth module to a smartphone. This guarantees that the user is in a position to monitor water quality levels in real time without having to personally collect it, thus lowering risk and enhancing efficiency.

The combination of turbidity sensor technology with a remotely operated boat is a modern and efficient method of water pollution monitoring. In contrast to traditional sampling techniques, this system minimizes the presence of man in toxic environments and enables data to be collected from several locations in the same body of water. It facilitates quicker and precise determination of areas of pollution hotspots and promotes improved environmental decision-making. The

project also creates awareness regarding the significance of water quality by enabling real-time, direct information to be utilized for remedial actions.

The water pollution monitoring RC boat with a turbidity sensor is a step in environmental protection in innovation. By integrating sensor technology with wireless communication and remote mobility, the project offers a cost-effective, efficient, and safe means of monitoring water body pollution levels. The system promotes sustainable water resource management and ensures protection for aquatic life and public health. By further enhancing the system, for example, by incorporating additional sensors or mapping capabilities, the design could be scaled up into a full-scale monitoring system.

1.1 THEORETICAL BACKGROUND

Water pollution is among the most challenging environmental concerns as a result of quick industrialization, urbanization, and population increase that have cast a huge burden on natural water bodies. Contaminants like suspended solids, industrial effluents, agricultural runoffs, and domestic wastes contaminate the water, rendering it unfit for both human consumption, aquatic organisms, and agricultural use. Conventional water quality monitoring techniques normally entail manual sampling of water bodies and subsequent laboratory analysis. Though accurate, such techniques are often costly, time-consuming, and not very effective in offering real-time data. Manual sampling in large, contaminated, or dangerous areas where human access is unsafe or challenging might also be impractical. These constraints have spurred the evolution of novel techniques that utilize contemporary technologies in water quality monitoring.

Turbidity, one of the key water pollution indicators, is a cloudiness or haziness in water due to the suspended particles, microorganisms, and impurities. High turbidity undermines light penetration, impacting aquatic plant growth, and tends to signal contamination that can pose dangers to humans as well as aquatic communities. Real-time monitoring of turbidity is hence crucial to knowing the condition of water bodies and avoiding environmental deterioration. Sensor technology is a pragmatic solution to this requirement as it makes possible the measurement of turbidity directly from water without recourse to laboratory processes.

Remotely controlled vehicles have been highlighted as a good method of transporting monitoring equipment to different points in an aquatic body. Remotely controlled boats are used as mobile platforms that move across ponds, lakes, or rivers with sensors to collect data. Microcontroller integration with turbidity sensors allows data automation, whereas wireless communication modules allow the user to transfer this data to a device in real time. In the current project, a Bluetooth module has been utilized not just to send sensor data but also to accept navigation commands from the user. The boat can be controlled in both forward and backward directions according to these commands, making possible focused monitoring in specific areas. This minimizes reliance on human effort, makes it safer, and opens up flexibility in approaching various areas of the water body.

The idea of exploiting embedded systems for environmental monitoring forms the core of this project. A microcontroller serves as the processing device that gathers input from the turbidity sensor and sends the information to the user using Bluetooth. Meanwhile, the microcontroller translates navigation instructions given by the user and directs the boat's motors in response. This two-way application illustrates how embedded systems, together with wireless communication and sensor technology, can be used to address actual-world environmental problems. Despite the current limitation to turbidity measurement, it builds a solid theoretical basis for further improvement. The project can be improved by adding other sensors like pH, temperature, and dissolved oxygen to give a complete water quality analysis or by adding GPS modules for geotagging the level of pollution in future. Accordingly, the theoretical framework of this research underscores the significance of uniting environmental science with contemporary electronics to develop inexpensive, reliable, and efficient ways of monitoring and managing water resources.

1.2 OBJECTIVE OF THE PROJECT

The main objective of this project is to design and implement a water pollution monitoring system using a remotely controlled boat equipped with a turbidity sensor. The system is intended to provide an efficient, accurate, and safe method for assessing water quality by measuring turbidity levels in real time. Since turbidity reflects the clarity of water and indicates the presence of suspended particles and pollutants, monitoring this parameter plays a vital role in evaluating the overall health of aquatic environments. By enabling direct data collection from different areas of a water body, the RC boat ensures that samples can be analyzed without the need for manual collection, which is often time-consuming, labor-intensive, and sometimes unsafe in contaminated zones. This reduces human effort, enhances safety, and ensures timely monitoring of water quality.

Another key objective is to integrate both navigation and data transmission using a Bluetooth module. This module allows the RC boat to be controlled in forward and backward directions by receiving commands from the user, while simultaneously transmitting turbidity readings to a mobile device in real time. The dual-purpose functionality of Bluetooth not only simplifies the system but also makes it more cost-effective by eliminating the need for multiple communication modules. This integration ensures that both movement and monitoring are achieved through a single channel, making the system compact, efficient, and user-friendly.

A further objective is to utilize readily available and low-cost components such as the Arduino Uno microcontroller, breadboard, DC motors, 9V batteries, and lightweight construction materials like thermocol and cardboard for building the boat structure. The use of simple components makes the project more affordable and replicable for academic, research, or small-scale field applications. This also emphasizes the project's focus on demonstrating how embedded systems and basic electronic components can be combined to create practical solutions for environmental issues.

Additionally, the project aims to provide a prototype that can serve as a foundation for future enhancements. By validating the working of a single-parameter monitoring system, it opens opportunities for integrating more advanced sensors such as pH, dissolved oxygen, or temperature sensors in subsequent versions. This scalability ensures that the project not only meets its current objective of turbidity monitoring but also highlights its potential for expansion into a more comprehensive water quality assessment system.

The overall goal of this project is to create a portable, reliable, and efficient water pollution monitoring solution that demonstrates the real-world application of embedded systems and wireless technology. Through its design, the project seeks to contribute to sustainable environmental management, promote awareness of water quality issues, and inspire further research in the field of smart monitoring systems. By addressing practical challenges such as safety, cost, and accessibility, the RC boat-based monitoring system stands as an innovative approach toward protecting water resources and ensuring healthier aquatic ecosystems for the future.

1.3 Purpose, Scope and Applicability

Purpose

The purpose of this project is to design and develop an efficient and affordable system for monitoring water pollution using a remotely controlled boat. Water pollution is a serious global issue that poses risks to both human health and aquatic life, and conventional monitoring techniques often require manual collection of samples followed by laboratory analysis. These traditional methods are not only time-consuming but also expose individuals to potential hazards when dealing with contaminated or unsafe water bodies. To overcome these challenges, the proposed system provides a safer and faster way of analyzing water quality directly from the field. By using a turbidity sensor integrated with an Arduino Uno microcontroller and supported by Bluetooth communication, the system delivers real-time data that reflects the clarity of water, thereby indicating the presence of suspended impurities and pollutants.

Scope

The scope of this project is focused on building a low-cost prototype that demonstrates the use of embedded systems and wireless technology in environmental monitoring. The RC boat is designed using thermocol and cardboard for light weight and stability, powered by two DC motors with fans and 9V batteries for mobility. The boat is capable of moving forward and backward through commands sent via Bluetooth, which also transmits turbidity data simultaneously to a mobile device. The system is limited to turbidity measurement, but its modular design ensures that it can be extended in future to include other sensors such as pH, dissolved oxygen, and temperature, thereby transforming it into a more comprehensive water quality monitoring solution. The project's scope also emphasizes cost-effectiveness and simplicity, making it suitable for educational demonstrations, small-scale monitoring tasks, and as a base model for advanced research.

Applicability

The applicability of this project lies in its practical use for monitoring pollution in lakes, rivers, ponds, and reservoirs. It can be deployed by environmental agencies, researchers, or even communities to quickly assess the condition of local water bodies. Since the system is portable and remote-controlled, it is particularly useful in situations where manual monitoring is difficult or unsafe, such as near industrial discharge points or contaminated zones. Moreover, the system highlights how technology-driven solutions can be applied to real-world environmental issues, serving as an awareness tool for sustainable water resource management. In academic settings, the project is applicable as a learning model that demonstrates the integration of hardware, software, and communication modules in solving societal problems. Overall, the purpose, scope, and applicability of this project underline its significance as an innovative step toward safer, smarter, and more accessible water quality monitoring systems.

2: Requirements and Analysis: -

2.1 PROBLEM DEFINATION

Water is one of the most essential natural resources that is needed for humans, animals, and plants to survive, as well as for industrial and agricultural purposes. However, due to the fast rate of industrialization and urbanization, water pollution has become a serious world problem. Spills of untreated industrial effluents, agricultural runoff with pesticides and fertilizers, and domestic wastes into rivers, lakes, and ponds have seriously contaminated water. It not only endangers aquatic life but also poses a severe threat to human health when polluted water is ingested or utilized for daily purposes. Of all the indicators of water pollution, turbidity is significant as it indicates the suspension of particles, sediments, and pollutants in water. High turbidity causes a decrease in light penetration, interferes with photosynthesis in water plants, and generally signifies contamination that can transport dangerous microorganisms or toxic chemicals. Real-time monitoring of turbidity levels is hence crucial to assess the magnitude of pollution and initiate preventive measures accordingly.

Traditional methods of monitoring water quality primarily depend on manual sampling and subsequent laboratory analysis. Though these approaches yield correct outcomes, they are very time-consuming, need expert labor, and entail significant costs. Furthermore, manual sampling may be inconvenienced in terms of frequency and geographical accessibility, thus unable to detect the fast-changing levels of pollution in different parts of a water body. In some situations, like polluted or remote areas, manual collection can also prove hazardous for people. These constraints call for complementary monitoring strategies that are safer, quicker, and can deliver real-time data to enable timely decision-making.

The challenge is in creating a system that will integrate mobility, safety, and efficiency with real-time monitoring. A stationary water monitoring station has no capacity to survey vast areas, and human-based sampling is limited by effort and safety considerations. Hence, a mobile platform is needed to transport sensors into varied areas of water bodies and supply continuous data without any human presence in risky areas. In solving this issue, the idea of remotely operated boat has been used in this project. The RC boat can be driven across a body of water while equipped with a turbidity sensor to take measurements in real-time. Data from the sensor is sent to a mobile device by a Bluetooth module, allowing the user to view pollution levels in real-time. The same Bluetooth

module receives driving commands to propel the boat in both forward and backward directions so that particular areas of interest can be reached cost-effectively.

Therefore, the problem solved by this project is the nonexistence of an economical, secure, and cost-effective system for real-time monitoring of water pollution. Through emphasis on turbidity measurement and the use of Bluetooth-based control and communication, the project aims to overcome the shortcomings of conventional methodologies while showing how embedded systems can deliver real-world solutions to environmental issues. While the present scope is only turbidity detection, the problem definition provides avenues for future enhancement by introducing additional sensors and technologies to enable complete water quality monitoring.

2.2 SOFTWARE REQUIREMENT SPECIFICATION (SRS)

In any system design, dividing the project into modules provides clarity, structure, and efficiency in implementation. A module can be defined as a logical component of the system that performs a specific function independently but also contributes to the overall working of the project. By breaking the project into smaller parts, it becomes easier to design, test, and debug each portion while ensuring that the integration of all modules results in smooth functioning of the complete system. In the project “Water Pollution Monitoring Using RC Boat,” six major modules have been identified: the Power Supply Module, Microcontroller Module, Bluetooth Communication Module, Sensing Module, Locomotion Module, and Boat Structure Module. Each of these modules performs a specific role and together they achieve the main objective of monitoring water pollution using a simple, cost-effective, and innovative approach.

Modules

1.Power Supply Module

The power supply module constitutes the core of the entire project, for any electronic system is worthless without a reliable source of power. In this project, two 9V batteries are employed to provide a guaranteed source so that all the components are supplied with the required supply for easy running. One battery is reserved for powering the Arduino Uno microcontroller and the turbidity sensor, while the other battery is reserved for driving the DC motors. The reason for this power source separation is that motors typically draw higher current and generate electrical noise, hence may cause fluctuations or interference in sensitive devices such as sensors and microcontrollers. By isolating the power lines, the system is made more stable and reliable. The breadboard provides the main platform where all the components are powered with the battery power to ensure neat, flexible, and easy-to-manage wiring for alterations. In the absence of a well-engineered power module, the performance of other modules would be affected, hence making this the most basic component of the system.

2. Microcontroller Module

The microcontroller module is the intelligence behind the project. For this purpose, the Arduino Uno is utilized because it is inexpensive, readily available, and has a friendly programming environment for beginners. The Arduino captures raw analog values from the turbidity sensor, processes and filters the readings, and sends them as useful data to the user via the Bluetooth module. At the same time, the Arduino gets movement command from the Bluetooth module and accordingly manages the DC motors to move the boat in the forward or backward direction. The Arduino's dual-tasking capability allows it to be a perfect candidate since it can easily monitor and control duties simultaneously. Additionally, the Arduino platform offers the flexibility of future upgrades, for instance, adding more sensors or GPS integration for enhanced capabilities. Therefore, the microcontroller module is the control center where all the information is processed and decision is taken, maintaining coordination among the rest of the modules.

3. Bluetooth Communication Module

One of the most significant parts of this project is wireless communication. The HC-05 Bluetooth module has been used due to its reliability, low price, and easy interfacing with Arduino Uno. This module facilitates two-way communication between the user's mobile device and the RC boat. It acts in two ways: first, it sends real-time turbidity sensor values from the Arduino to the user's mobile device, enabling the user to track water quality in real time. Second, it receives input commands from the user, e.g., forward and reverse movement, and relays them to the Arduino for implementation. This two-way application of Bluetooth obviates the necessity for more than one communication module, making the system simpler while maintaining the cost-effectiveness of the system. Bluetooth also ensures that the communication process is wireless, thus enhancing safety, since the user is not required to be in close contact with contaminated water while monitoring. This module illustrates how wireless communication technologies can make embedded systems more efficient and easy to use.

4. Sensing Module

At the center of the project is the sensing module, which comprises the turbidity sensor. Turbidity is one of the major parameters used in measuring water quality since it reflects the concentration of suspended solids, sediments, and contaminants in water. The turbidity sensor operates by detecting the light intensity that passes through the water sample: pure water lets more light pass, while dirty water with greater turbidity scatters or absorbs the light, causing varied readings. The readings are transmitted to the Arduino Uno as analog signals, which are calculated and sent to the user via Bluetooth. The instant measurement of this sensor makes it possible to immediately sense the level of pollution, in contrast to the usual laboratory-type systems with time lags. Because turbidity impacts water organisms directly and is usually indicative of contamination, this module plays an essential role in achieving the prime purpose of monitoring pollution. It is also the most research-based module, since the data gathered form the foundation of determining the condition of water bodies.

5. Locomotion Module

The locomotion module is assigned with the movement of the boat physically on the surface of water. Two DC motors are employed as propelling units within this project. They are attached to the Arduino and supplied with a dedicated 9V battery for independent working without interfering with other units. The Arduino manages the motors based on instructions from the Bluetooth module. The ability of the boat to move forward and backward enables the operator to steer it towards various areas of the water body for focused monitoring. Despite its limited two directions of travel, this module offers the flexibility needed for small- and medium-scale water monitoring. The locomotion module not only guarantees mobility but also shows how mechanical systems can be coupled with embedded electronics to build functional robotic systems. This mobility provides immense value, since immovable sensors are unable to monitor greater expanses of a water body, whereas a moving platform can sample more representative information.

6. Boat Structure Module

The boat structure module gives the physical body and housing for everything. In this project, the boat is constructed using thermocol and cardboard, which are light in weight, cheap, and easily available materials. Thermocol ensures that the boat stays afloat without sinking, while cardboard adds support and structural integrity for mounting elements such as the Arduino Uno, Bluetooth module, breadboard, batteries, turbidity sensor, and DC motors. Component organization is done carefully to achieve a balance between weight and to avert tilting of the boat when running. Although constructed from low-cost materials, the boat structure is strong enough for use in small-scale monitoring in rivers, lakes, and ponds. The boat design shows that functionality and practicality can still be achieved from low-cost design strategies. This module finishes integrating the electronics and mechanics into one mobile system, making it possible for the project to function in real-world water conditions.

Functional Requirements

Functional requirements define what the system must do and describe the core features necessary for its proper operation. The following functional requirements are identified for this project:

- **Water Quality Monitoring:** The system shall be capable of measuring the turbidity of water using the turbidity sensor. The sensor must detect the cloudiness or suspended particles in the water and send an analog signal to the Arduino Uno. This data should then be processed to provide a clear indication of whether the water is clean or polluted.
- **Data Transmission:** The system shall transmit the processed turbidity data to the user in real time. The HC-05 Bluetooth module will handle wireless communication, ensuring that the values collected from the sensor are sent instantly to a paired mobile device for monitoring.
- **Movement Control:** The system shall provide the user with the ability to control the RC boat's movement through Bluetooth commands. The boat should be able to move forward and backward smoothly, allowing the user to navigate it into different areas of a water body for targeted testing.

- **Microcontroller Processing:** The Arduino Uno shall act as the brain of the system by processing input from the turbidity sensor, interpreting Bluetooth commands, and sending appropriate signals to the DC motors. This ensures that both water quality measurement and movement operations are handled efficiently.
- **Power Management:** The system shall make use of two 9V batteries, one dedicated to powering the Arduino and sensor, and the other for the motors. This separation is necessary to ensure that motor power fluctuations do not interfere with sensor readings or microcontroller operations.
- **User Interaction:** The system shall provide a simple interface for the user, where commands can be sent through a Bluetooth-enabled mobile device. The user should also be able to view turbidity readings directly on the device without the need for complex setup or programming knowledge.
- **Real-Time Operation:** The system shall operate in real time. This means that when the user issues a navigation command, the boat should respond immediately. Similarly, when the boat collects water quality data, the values should be sent without delay to the mobile device.

Non-Functional Requirements

Non-functional requirements describe the overall qualities and constraints of the system, ensuring reliability, usability, and efficiency. The following non-functional requirements are considered for this project:

- **Performance:** The system should deliver accurate and reliable turbidity measurements within a few seconds. Both the movement and data transmission should happen without noticeable delay, ensuring smooth operation.
- **Reliability:** The RC boat should be able to operate consistently under normal conditions without frequent breakdowns. It must remain stable in water and continue functioning even when exposed to small disturbances such as ripples or minor water flow.

- **Usability:** The system should be user-friendly and not require any advanced technical expertise to operate. A simple Bluetooth connection between the RC boat and a mobile device should be sufficient to control movement and receive pollution data.

- **Portability:** The boat should be lightweight and compact, making it easy to carry to lakes, ponds, or rivers. The use of thermocol and cardboard ensures that the boat remains portable while still being strong enough to hold the electronic components.

- **Cost-Effectiveness:** The project should be designed using low-cost and readily available components such as Arduino Uno, HC-05 Bluetooth module, DC motors, and thermocol. This keeps the overall budget affordable, making it suitable for student projects and small-scale environmental monitoring.

- **Maintainability:** The design of the boat should allow easy maintenance. Components such as sensors, motors, or batteries should be easily replaceable without requiring complete rebuilding of the boat. This makes the system durable and long-lasting.

- **Scalability:** The system should be designed in such a way that additional sensors such as pH or temperature sensors can be added in future upgrades without major changes to the hardware. This ensures that the project has scope for future development.

- **Safety:** The system should ensure the safety of its electronic components by providing proper insulation and protection against water damage. The use of lightweight, non-toxic materials such as thermocol also makes the boat safe to handle and environmentally friendly.

2.3 FEASIBILITY STUDY

A feasibility study is done to assess the practicability and viability of a project prior to its complete development and implementation. It aids in analyzing the strengths, weaknesses, opportunities, and potential challenges of the system being under consideration. The study also identifies whether the system to be put in place can be successfully developed using the available resources, technical competencies, financial limits, and operation environment. For the project Water Pollution Monitoring Using RC Boat, feasibility analysis has been conducted under various categories, such as technical feasibility, economic feasibility, operational feasibility, and social feasibility.

Technical Feasibility

Technical feasibility tests whether the project can be designed and executed based on the available technology, tools, and resources. In this project, the major components are an Arduino Uno microcontroller, HC-05 Bluetooth module, turbidity sensor, DC motors, batteries, and boat structure using thermocol and cardboard. All these components are affordable, readily available, and compatible with each other, which makes the technical part greatly achievable. Arduino Uno is selected due to its ease of use, large support base, and capability to simultaneously process sensor data and motor control. HC-05 Bluetooth module ensures stable short-distance communication for smooth turbidity data transmission and navigation commands. The turbidity sensor efficiently detects suspended material in water, which is the primary parameter to identify water pollution. In addition, the application of DC motors guarantees smooth travel of the boat over the surface of the water. The implementation of these units demands minimal knowledge of embedded systems, and the design can be accomplished with existing laboratory equipment. The project, therefore, is technically feasible.

Economic Feasibility

Economic feasibility considers if the project is economically viable and if the advantages overrule the expenses. The project implements low-cost modules like the Arduino Uno, turbidity sensor, HC-05 Bluetooth module, and easy-to-obtain construction materials like thermocol and cardboard. Development cost is low relative to commercially purchased water quality monitoring setups,

which tend to be costly and complex in terms of maintenance. The setup can also be reused several times with little maintenance costs incurred, such as changing batteries or replacing minor mechanical components. The cost-effectiveness of the design allows it to be affordable for students, researchers, and small-scale organizations that cannot afford huge systems. Given its low investment and high utilization in environmental monitoring, the project is economically viable.

Operational Feasibility

Operational feasibility revolves around whether the project is usable and whether the target users are able to use it with minimal difficulty. The RC boat is easy to use and only needs a Bluetooth-enabled phone to control and monitor data. The system does not require specialized technical skills to operate since the boat can be propelled forward and backward with basic commands, and water quality readings are accessed in real-time on the linked device. The configuration of the boat provides stability in water and shields the mounted components, thus minimizing the risk of malfunctioning when running. Because the boat is light in weight and portable, it can easily be used in lakes, ponds, rivers, and other bodies of water. Therefore, the system is very practical for application in real life, and operational feasibility is well established.

Social Feasibility

Social feasibility assesses the extent to which the project integrates within its targeted social and environmental setting. Pollution of water is a severe problem that impacts human health, aquatic organisms, and the long-term use of natural resources. The intended project makes a positive contribution through the provision of a secure, low-cost, and creative means of monitoring the quality of water. It reduces the risks associated with sample collection manually, particularly in contaminated or unsafe water sources, thus safeguarding human health. In addition, the project increases public sensitivity regarding the need for water quality checks and encourages prudent application of technology for safeguarding the environment. Through real-time insights into pollution intensity, the system will help local authorities, environmentalists, and scientists in taking preventive and remedial steps. The social acceptability of this project is thus high, as it is addressing a critical environmental issue directly and promoting sustainable lifestyles.

2.4 DETAIL OF HARDWARE AND SOFTWARE USED

In the development of the project Water Pollution Monitoring Using RC Boat, both hardware and software elements have been integrated to achieve efficient functioning. Hardware is responsible for the structural framework and the execution of physical tasks, while software handles logic, communication, and overall system coordination. Together, they establish a practical and innovative solution to monitor water pollution, particularly by assessing turbidity in water bodies. This section describes in detail the hardware and software components employed in the system and their respective functions.

Hardware Parts

Arduino Uno

The Arduino Uno serves as the central control unit or the “brain” of the project. It is a microcontroller board based on the ATmega328P, which is widely used in embedded system applications due to its simplicity and efficiency. In this project, the Arduino reads signals from the turbidity sensor, interprets the data, and manages outputs accordingly. It also handles commands received via the Bluetooth module to control the motors. Its versatile input/output pins, built-in analog-to-digital converters, and compatibility with programming libraries make it the most suitable choice for this project. The Arduino ensures that all operations, such as motor control and sensor data acquisition, occur seamlessly and in real time.

Breadboard

The breadboard is a prototyping tool that allows components to be connected without soldering. It plays a critical role in simplifying circuit design by making it possible to connect, disconnect, and rearrange wiring easily during development. In this project, the breadboard is used to interconnect the Arduino, Bluetooth module, turbidity sensor, motor driver, and other components. It also facilitates the distribution of power from the 9V batteries to the respective modules. The flexibility offered by the breadboard significantly reduces errors and speeds up the process of building and testing the system.

HC-05 Bluetooth Module

The HC-05 Bluetooth module is the communication backbone of the system. It establishes a wireless link between the RC boat and the user's mobile device. Through this module, turbidity readings are sent to the user's phone, and navigation commands are received. The HC-05 is known for its stability and low power consumption, which makes it suitable for embedded applications. In this project, it performs dual roles—facilitating the transmission of sensor data and enabling navigation of the boat in forward and backward directions. This dual purpose minimizes the need for additional communication hardware, keeping the design both efficient and cost-effective.

Turbidity Sensor

The turbidity sensor is the primary sensor used in this system to evaluate the quality of water. It works by passing light through the water sample and measuring the amount of light scattered by suspended particles. A higher turbidity value indicates a greater concentration of pollutants or suspended solids. The sensor sends an analog signal to the Arduino, which is then processed and communicated to the user's device. This sensor is significant because turbidity is one of the most reliable indicators of water pollution and can provide immediate information about the level of contamination.

L9110S Motor Driver

The L9110S motor driver is a crucial component responsible for driving the two DC motors used in the boat. Arduino pins cannot directly supply sufficient current to run motors, which is why a motor driver is necessary. The L9110S provides the required amplification and ensures safe operation of the motors. It receives control signals from the Arduino and translates them into appropriate voltage and current levels for the motors. In this project, the motor driver controls the forward and backward motion of the boat, enabling smooth navigation in water. Its compact design and efficiency make it ideal for small-scale robotics projects such as this one.

Two 9V Batteries

The project employs two 9V batteries to supply power. One battery powers the Arduino Uno and turbidity sensor, ensuring steady operation of the control and sensing modules. The other battery is dedicated to the motor driver and DC motors, which prevents power fluctuations caused by motors from affecting the sensitive electronics. This separation of power sources contributes to stable performance and extends the lifespan of the components.

DC Motors with Propeller Fans

Two DC motors are installed in the propulsion system of the RC boat. They are connected to propeller fans that generate thrust when rotated. By controlling the rotation direction, the boat can move forward or backward as instructed by the user via Bluetooth. The lightweight design of the fans ensures efficient movement even in small water bodies. These motors are reliable, compact, and cost-effective, making them suitable for a student-level embedded project.

Switch

A switch is used in the RC boat to control the main power supply from the battery to the circuit. It acts as an ON/OFF controller, ensuring that the system only operates when required. This helps in conserving battery life, preventing unwanted power drain, and providing a safe way to start or stop the boat's operations.

Boat Structure (Thermocol and Cardboard)

The physical body of the boat is crafted using thermocol and cardboard. Thermocol is chosen for its buoyant property, ensuring that the boat floats reliably on water. Cardboard provides structural support, offering a base for mounting the Arduino, motors, breadboard, and other hardware. This combination results in a simple yet effective boat structure that is lightweight, low-cost, and easy to assemble. Though basic in design, the structure meets the functional requirements of the project and can be upgraded in future versions for greater durability.

Software Components

Arduino IDE

The Arduino Integrated Development Environment (IDE) is the software platform used to write, compile, and upload code to the Arduino Uno. It provides a user-friendly interface, preloaded libraries, and debugging tools, which simplify programming tasks. In this project, the Arduino IDE is essential for creating the logic required to read turbidity values, handle Bluetooth commands, and control the motor driver. The IDE also facilitates quick testing and iterative improvements during development.

Embedded C Programming

The logic for the Arduino Uno is written in Embedded C, which is widely used for programming microcontrollers. Embedded C allows direct access to hardware-level features, making it ideal for controlling sensors, motors, and communication modules. The program specifies instructions for data acquisition from the turbidity sensor, interpreting user commands through Bluetooth, and managing motor operations. This code acts as the bridge between hardware functionality and user interaction, ensuring real-time response and reliable operation.

Bluetooth Terminal/Mobile Application

A mobile device with a Bluetooth terminal application is used to interface with the RC boat. The application enables the user to send navigation commands (forward or backward) and simultaneously receive turbidity readings from the boat. This software component is vital for providing real-time interaction with the system, making it both interactive and user-friendly. Since smartphones are commonly available, this design choice ensures that the system can be operated without the need for specialized hardware or complex setup.

2.5 Planning and Scheduling

Project Planning

Project planning is a crucial step in the successful execution of any engineering or embedded system project. For the Water Pollution Monitoring RC Boat Project, careful planning ensures that all tasks are executed efficiently, resources are properly utilized, and the final objectives are achieved within the given timeframe. The main goal of this project is to design and develop a small, remotely controlled boat capable of measuring water pollution levels using a turbidity sensor and transmitting data to a mobile device via Bluetooth.

The planning phase begins with defining the project objectives. These include building a stable RC boat, integrating sensors and motor drivers, ensuring wireless communication through Bluetooth, and designing a compact and functional system. Next, resource identification is performed, including procuring the Arduino Uno, L9110 motor driver, DC motors, HC-05 Bluetooth module, turbidity sensor, batteries, rocker switch, and supporting components like breadboards, wires, and the boat chassis.

Following resource allocation, task breakdown is performed to divide the project into smaller, manageable modules. These modules include: (1) boat chassis preparation, (2) motor and motor driver integration, (3) sensor installation, (4) Arduino programming for motor control and sensor data acquisition, (5) Bluetooth communication setup, and (6) testing and calibration. This modular approach allows for parallel work on different sections and ensures systematic progress.

Risk management is also part of planning. Potential risks, such as component failure, battery drainage, Bluetooth connectivity issues, or sensor inaccuracies, are anticipated and mitigation strategies are developed. For example, spare components are kept ready, and calibration routines are designed to maintain sensor accuracy.

Lastly, planning involves setting milestones and checkpoints to monitor progress and ensure timely completion. Milestones such as completing hardware assembly, finishing coding, successful integration, and field testing are predefined. Regular evaluation and adjustments in the plan are carried out to address any delays or unexpected issues.

In conclusion, meticulous project planning lays the foundation for effective execution, ensuring that the Water Pollution Monitoring RC Boat Project meets its objectives, functions reliably, and provides meaningful environmental data.

Project Scheduling

Project scheduling defines the timeline and sequence of tasks to ensure the project is completed efficiently within the allocated period. For the Water Pollution Monitoring RC Boat Project, scheduling involves arranging tasks in a logical order, assigning durations, and establishing deadlines to maintain steady progress.

The first step in scheduling is task identification and sequencing. The main tasks include procurement of components, assembly of the boat chassis, installation of motors and motor driver, mounting the turbidity sensor, wiring and circuit setup on the breadboard, programming the Arduino for motor control and sensor readings, Bluetooth module integration, and system testing. Each task is arranged sequentially where dependencies exist, ensuring that one task's completion allows the next to proceed.

Next, time estimation is performed for each task. For example, component procurement may take 3–5 days, chassis preparation 2 days, motor and driver installation 1–2 days, sensor setup 1 day, coding 4–5 days, Bluetooth integration 2 days, and testing 2–3 days. Buffer time is also included to accommodate unforeseen delays.

A Gantt chart or timeline is often prepared to visualize the schedule. This chart highlights task start and end dates, overlapping tasks that can be done in parallel, and critical milestones. For instance, while the chassis is being prepared, programming of the Arduino can begin simultaneously to optimize time. Key milestones include completion of hardware assembly, successful coding of motor control, sensor calibration, and final field testing.

Monitoring and adjustment are integral to scheduling. Regular reviews of progress ensure that tasks are completed on time. Delays in one task may require rescheduling subsequent tasks or reallocating resources. Scheduling also includes planning for maintenance, calibration, and troubleshooting during the project's execution.

2.6 Preliminary Product Description

The Water Pollution Monitoring RC Boat is an embedded system designed for environmental monitoring of small water bodies such as ponds, lakes, and slow-moving rivers. The primary purpose of this system is to assess water quality by measuring turbidity levels, which indicate the presence of suspended particles and potential pollutants. The RC boat integrates mechanical, electrical, and software components to form a compact and functional monitoring platform.

The system is built around an Arduino Uno microcontroller, which serves as the brain of the project. It processes data from the turbidity sensor, controls the DC motors through the L9110 motor driver, and communicates wirelessly with a mobile device using the HC-05 Bluetooth module. The Arduino allows precise control over the boat's movement, enabling it to move forward, backward, or stop, while simultaneously acquiring water quality data.

The mechanical design consists of a lightweight RC boat chassis, on which the motors, propellers, sensors, and electronic components are mounted. The compact design ensures stability and ease of navigation in calm water environments. A rocker switch is included to provide manual control for powering the system on or off safely. Power is supplied through 9V batteries, providing sufficient energy for the Arduino, motors, and sensors during short-term field operations.

The boat's primary functional capabilities include real-time measurement of water turbidity, wireless transmission of data to a mobile device, and remote-controlled navigation. The turbidity sensor allows the system to detect water pollution, providing a simple yet effective tool for environmental monitoring. The Bluetooth module ensures that data can be monitored and recorded in real time without physical connection, making the system user-friendly and mobile.

In terms of applications, this RC boat can be used by students, researchers, and environmental organizations to monitor water quality in local water bodies. It provides a low-cost, portable, and educational solution for understanding water pollution trends. The system also allows scope for future upgrades, such as adding additional sensors for pH, temperature, or chemical contaminants, and implementing autonomous navigation for extended monitoring.

3: System Analysis and Design:-

3.1 DETAILED LIFE CYCLE OF THE PROJECT

Our project follows the **Waterfall Model** of SDLC. This model is chosen because the requirements of the system were well-defined from the beginning and did not undergo frequent changes. Unlike the Incremental Model, where the system is built and delivered in multiple partial increments, our project was designed and implemented as a complete integrated system. Since the boat's hardware and software components needed to be developed and tested together, a sequential approach provided by the Waterfall Model was the most suitable.

1. Requirement Analysis

Requirement analysis was the initial project lifecycle stage. During this phase, all the required hardware and software requirements were determined after a thorough examination of the issue of monitoring water pollution. The key requirements were Arduino Uno microcontroller, HC-05 Bluetooth module for wireless connectivity, turbidity sensor to detect water quality, power supply through two 9V batteries, DC motors to move the boat, breadboard for interfacing, and building materials like thermocol and cardboard to create the boat. Software requirements comprised the Arduino IDE for coding and serial observation. The specifications were completed to guarantee that the boat would be able to detect water turbidity, transmit data over Bluetooth to a cell phone, and travel forward and backward at the control of the user.

2. System Design

Once the requirements were outlined, the system design was conducted. The design consisted of both hardware and software design. On the hardware front, the location of Arduino Uno, breadboard, motors, and sensor was determined to provide balance and stability to the boat. The HC-05 module was located for stable wireless transmission. The body of the boat was created with thermocol for buoyancy and cardboard for mounting and support of components. On the software front, the operation flow was planned, such as how the Arduino would read turbidity sensor readings, process them, and send them to the mobile app through Bluetooth. The logic for the forward and backward movements with motor driver connections was also designed. This phase guaranteed a clean outline before proceeding to implementation.

3. Hardware Development

On this stage, the hardware parts of the system were constructed. The thermocol base was trimmed and molded to create the shape of the boat, whereas cardboard sheets were employed for mounting the electronics. The Arduino Uno, HC-05, turbidity sensor, motors, and batteries were mounted on the boat with caution. Electrical connections were made with a breadboard and jumper wires. Motors were placed such that there is smooth forward and backward movement of the boat in water. Particular care was taken to waterproof and make sure that wires and components were properly secured so that they would not get damaged during use.

4. Software Development

After the hardware was prepared, the Arduino Uno was programmed using the Arduino IDE. The program was designed to read turbidity sensor values, compare these to thresholds, and transmit the information to an associated mobile device via the HC-05 Bluetooth module. More code was added to permit user commands for the forward and reverse motion of the boat through the control of DC motors. Code testing was performed first on a breadboard using basic LEDs and motors before finalizing the integration. This allowed sensing and movement functions to be tested as desired.

5. Integration and Testing

Once hardware and software development, the integration process was executed. All the components were interfaced and powered to test if they could function together as a whole system. The turbidity sensor was checked in clean and contaminated water samples for its accuracy to measure the water quality. Bluetooth communication was also checked by pairing HC-05 with a mobile phone to ensure error-free data transfer. In the same way, the boat's forward and backward movement was tested in varying conditions to assess motor performance as well as battery efficiency. Debugging was performed to correct minor faults in connections and code to make the boat run smoothly.

6. Deployment

After testing was accomplished, the project was implemented for real-world use. The RC boat was taken to larger water bodies to test its actual functionality. The deployment confirmed that the system could measure turbidity levels properly and enable movement control via Bluetooth commands. The deployment process also ensured the practical applicability of the project to monitor pollution in ponds, lakes, or small rivers.

7. Maintenance and Future Enhancement

The last phase of the lifecycle is maintenance. Despite deployment, the system needed periodical maintenance like recharging or battery replacement, tying loose wires, and recalibration of turbidity sensor for precision. Enhancements in future include installing more sensors for pH and dissolved oxygen measurements, GPS integration for mapping areas contaminated, and data storage on a cloud platform for long-term analysis. These improvements can make the system more advanced and useful for large-scale water pollution monitoring.

3.2 Context Diagram

A System Context Diagram (SCD) is a high-level representation that shows the system as a single process and its interaction with external entities. In the Water Pollution Monitoring Using RC Boat project, the context diagram highlights how the user communicates with the RC boat through a mobile device using Bluetooth, while the turbidity sensor provides water quality data that is processed by the Arduino and sent back to the user.

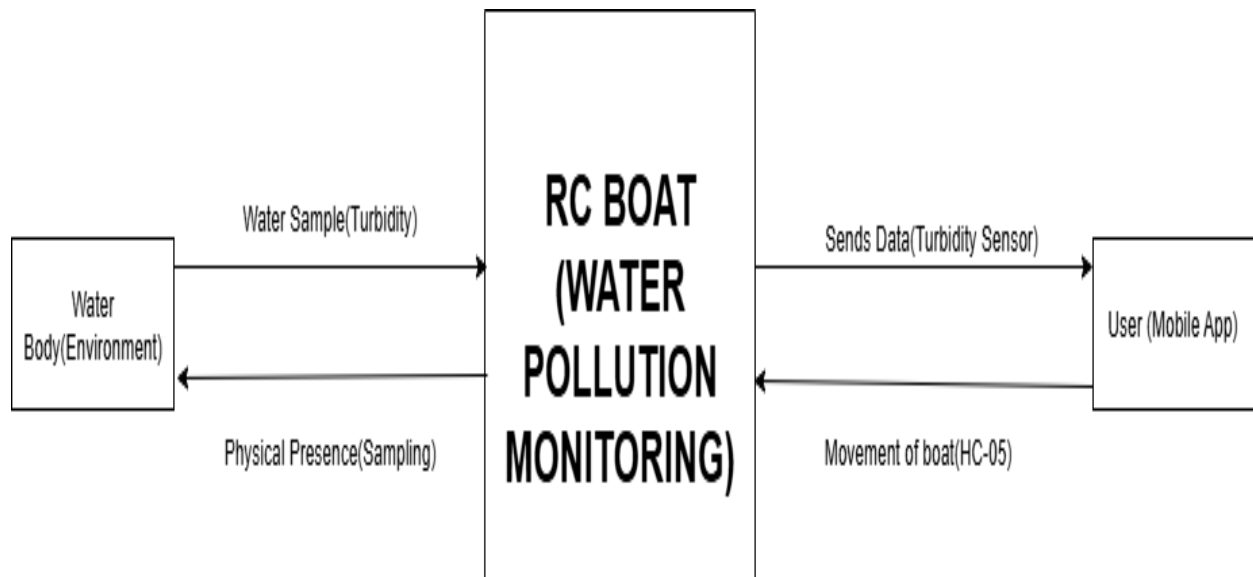


Fig 3.2: Context diagram

3.3 Data Flow Diagram

A Data Flow Diagram (DFD) is used to represent the flow of data within a system. In the Water Pollution Monitoring Using RC Boat project, the DFD shows how turbidity data is collected by the sensor, processed by the Arduino, and sent to the user's mobile device via Bluetooth, along with how user commands control the boat's movement.

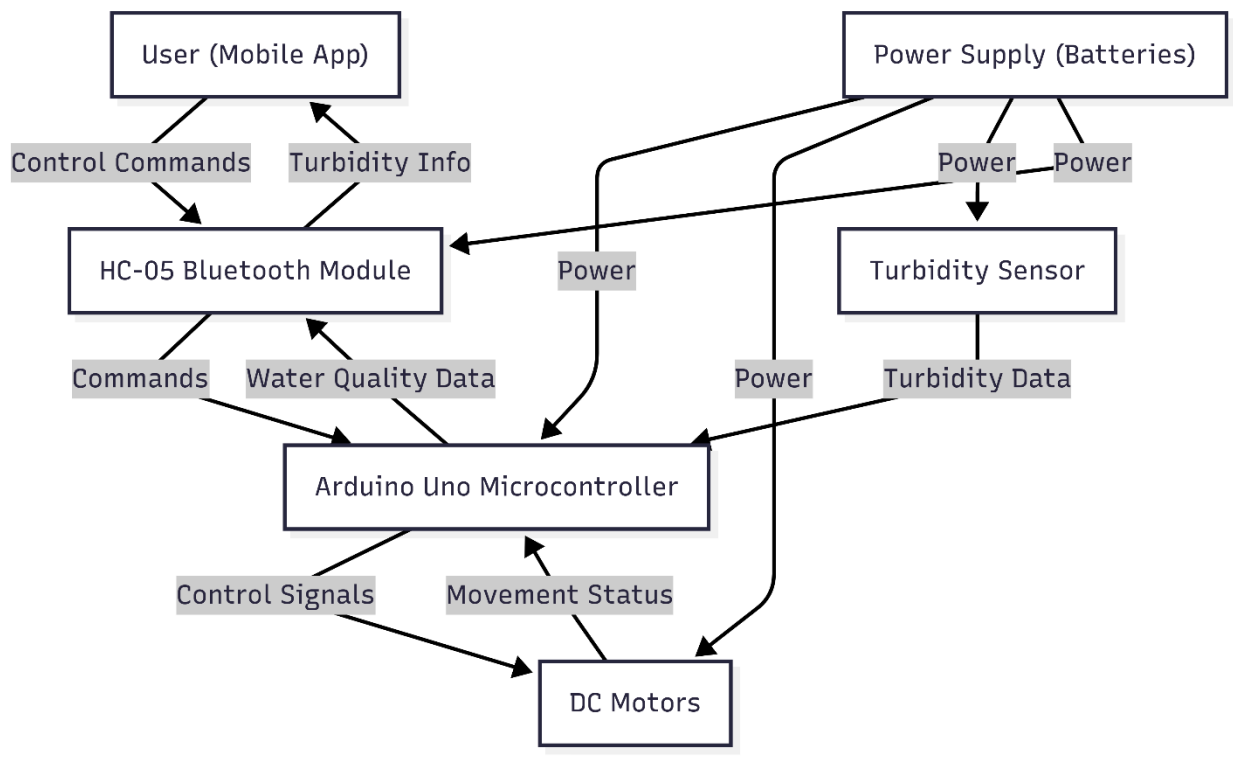


Fig 3.3: Data flow Diagram

3.4 State Transition Diagram

A State Transition Diagram represents the different states of a system and how it changes from one state to another based on events or inputs. In the Water Pollution Monitoring Using RC Boat project, it shows the states of the boat such as idle, moving forward, moving backward, and sending sensor data, along with the transitions triggered by user commands and sensor readings.

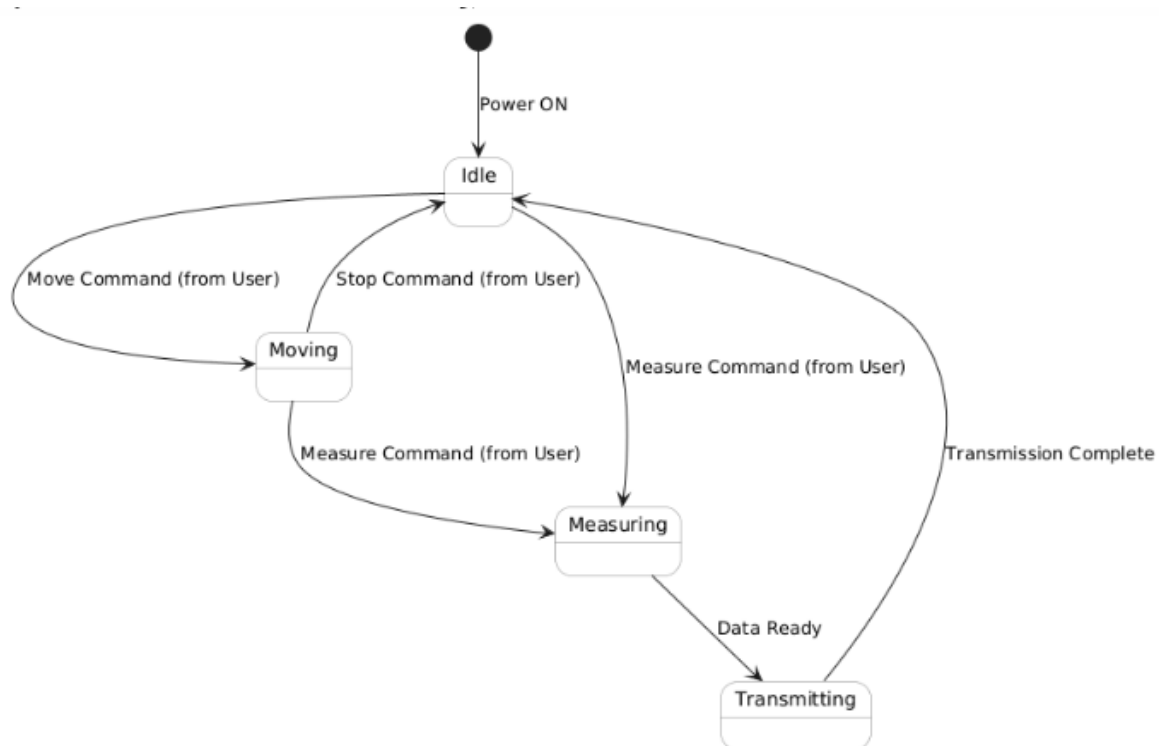


Fig 3.4: State Transition Diagram

3.5 Use Case Diagram

A Use Case Diagram illustrates the interactions between users and the system, showing the different functions the system provides. In the Water Pollution Monitoring Using RC Boat project, the use case diagram highlights how the user can control the boat's movement through Bluetooth and monitor water quality data received from the turbidity sensor.

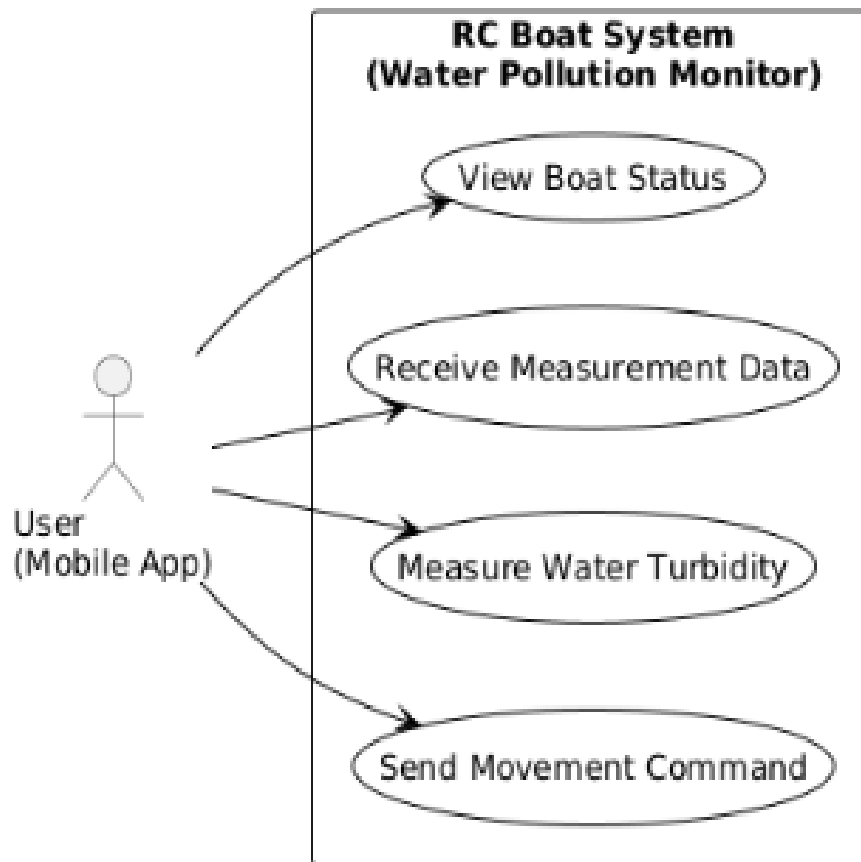


Fig 3.5: Use Case Diagram

3.6 Circuit Diagram

The circuit diagram illustrates the connections between the main components of the RC Boat project. The Arduino Uno serves as the central controller, connected to the turbidity sensor for water quality measurement, the HC-05 Bluetooth module for wireless control, and the L9110 motor driver to operate the two DC motors. Power is supplied through two 9V batteries, with one dedicated to the motors and the other to the controller and sensors. The breadboard is used for easy wiring and prototyping. This diagram provides a clear overview of how signals and power flow across the system.

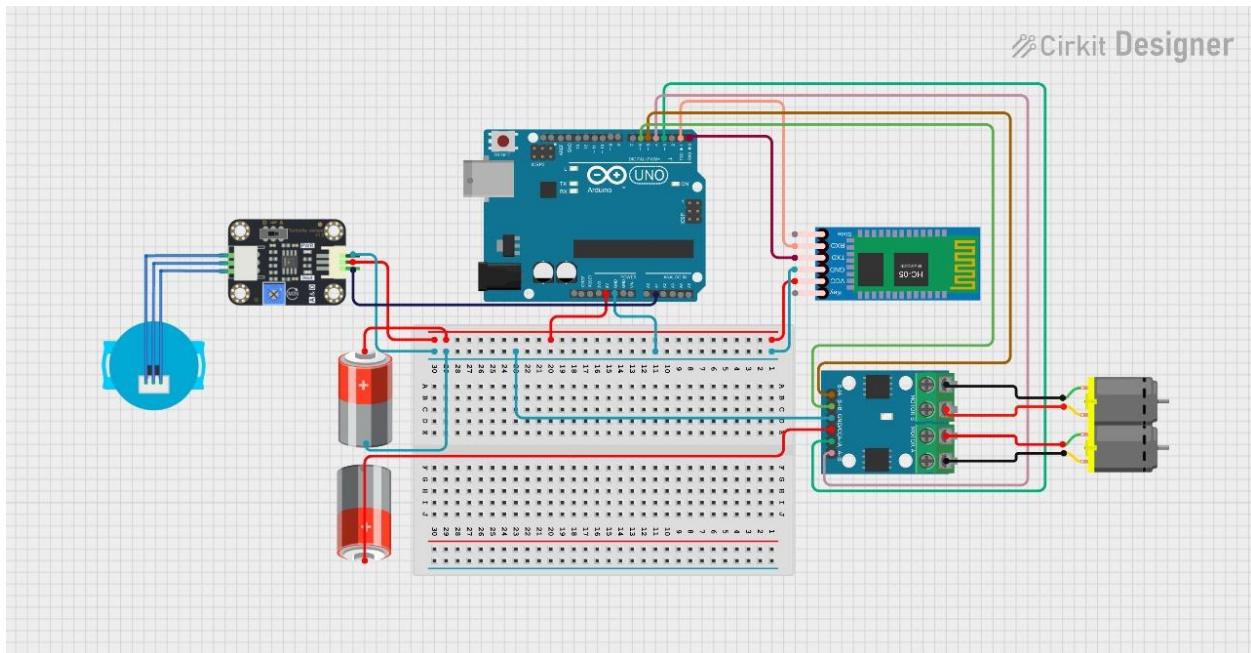


Fig 3.6: Circuit Diagram

Circuit Diagram Explanation

The circuit diagram illustrates the complete hardware design of the Water Pollution Monitoring System using RC Boat. The system integrates various components such as the Arduino Uno, HC-05 Bluetooth module, turbidity sensor, L9110 motor driver, DC motors, batteries, and a breadboard to create a functional and efficient setup.

At the core of the circuit is the **Arduino Uno**, which serves as the central controller. It receives input signals from the turbidity sensor, processes the data, and then transmits it to the mobile device via the Bluetooth module. The Arduino also controls the movement of the boat by sending signals to the motor driver based on the navigation commands received from the user through Bluetooth.

The **turbidity sensor** is connected to the Arduino to measure water quality. It provides an analog output corresponding to the cloudiness of the water. This reading is processed by the Arduino and later transmitted wirelessly.

The **HC-05 Bluetooth module** is connected to the Arduino's TX and RX pins to facilitate wireless communication. It performs a dual role: sending the turbidity readings to the mobile device and receiving navigation commands from the user.

For locomotion, the **L9110 motor driver** is used to interface the Arduino with the two DC motors. The motor driver acts as a bridge, allowing low-power signals from the Arduino to control the higher power required by the motors. Each DC motor is connected to the motor driver outputs and mounted with fans/propellers to enable forward and backward movement of the boat.

The system is powered using **two 9V batteries**. One battery supplies stable power to the Arduino and turbidity sensor, while the other is dedicated to running the DC motors via the motor driver. This separation ensures that fluctuations from the motors do not affect the sensor readings or Arduino performance.

The **breadboard** is used to simplify the connections between the components. It allows easy wiring, distribution of power lines, and flexibility during testing and modification.

3.7 Component Level description and specifications

In this project, several hardware and software components were integrated to form a functional water pollution monitoring system using an RC boat. Each component plays a significant role in ensuring that the system operates efficiently. The following section provides a detailed description and specifications of the components used.



Fig 1: Arduino Uno

1. Arduino Uno

Description: Arduino Uno is the central microcontroller board used in this project. It acts as the brain of the system, controlling sensor data collection, motor operations, and communication with the Bluetooth module. It is based on the ATmega328P microcontroller and provides multiple digital and analog pins for interfacing.

Specifications:

- Microcontroller: ATmega328P
- Operating Voltage: 5V
- Input Voltage (recommended): 7–12V
- Digital I/O Pins: 14 (6 PWM outputs)
- Analog Input Pins: 6
- Clock Speed: 16 MHz

- Memory: 32 KB Flash, 2 KB SRAM

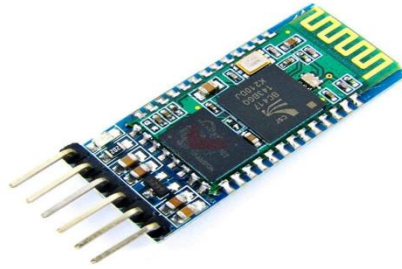


Fig 2: HC-05 Bluetooth Module

2. HC-05 Bluetooth Module

Description: The HC-05 Bluetooth module enables wireless communication between the RC boat and the user's mobile device. It is used both for transmitting turbidity sensor readings and for receiving control commands (forward and backward movement).

Specifications:

- Communication: Serial (UART)
- Operating Voltage: 3.3–5V
- Bluetooth Protocol: v2.0+EDR
- Range: 10 meters (approx.)
- Default Baud Rate: 9600 bps
- Mode: Master/Slave configurable



Fig 3: Turbidity Sensor

3. Turbidity Sensor

Description: The turbidity sensor measures the cloudiness of water, which is directly related to pollution levels. It detects the presence of suspended particles in water and provides analog values to the Arduino for processing.

Specifications:

- Operating Voltage: 5V DC
- Measuring Range: 0–1000 NTU
- Output: Analog (to Arduino ADC pin)
- Current Consumption: < 30 mA
- Response Time: < 500 ms



Fig 4: DC Motors with Propeller Fans

4. DC Motors with Propeller Fans (2 Units)

Description:

Two DC motors are used to drive the movement of the RC boat. Each motor is attached to a small plastic propeller (fan) that rotates in water to generate thrust. By controlling the rotation direction of the motors, the boat can move forward, backward, or turn. This setup provides both propulsion and steering without the need for a separate rudder.

Specifications:

- Operating Voltage: 3–12V DC
- Rated Speed: 1000–3000 RPM (depending on applied voltage)
- Current Consumption: 100–250 mA per motor
- Direction: Bidirectional (forward/reverse using polarity change)
- Propellers: Plastic propeller fans (diameter ~5–7 cm)
- Function: Converts motor rotation into thrust for boat movement

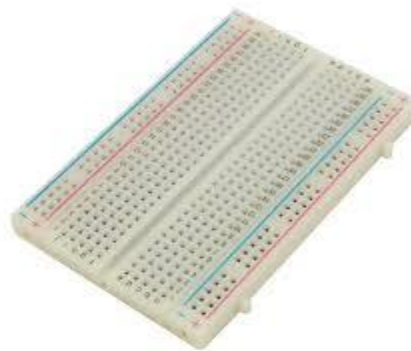


Fig 5: Breadboard

5. Breadboard

Description: A breadboard is used for connecting electronic components without soldering. It allowed flexible wiring of the Arduino, Bluetooth module, turbidity sensor, and motors during

prototyping.

Specifications:

- Type: 400–830 tie-points
- Material: ABS plastic with metal clips
- Connection: Spring-loaded contacts
- Reusability: Yes



Fig 6: 9v Battery

6. Power Supply (Two 9V Batteries)

Description: Two 9V batteries are used as the power source for the boat, supplying energy to the Arduino, sensors, and motors. Proper distribution ensures uninterrupted operation of both sensing and movement functions.

Specifications:

- Type: Alkaline 9V batteries
- Nominal Voltage: 9V
- Capacity: 400–600 mAh
- Weight: ~45 g each



Fig 7: L9110 Motor Driver

7.L9110 Motor Driver

Description:

The L9110 motor driver is a compact dual-channel H-bridge module used to control two DC motors simultaneously. In our water pollution monitoring RC boat project, it is responsible for driving the two DC motors that move the boat forward and backward based on commands received from the Arduino Uno via the Bluetooth module. The L9110 acts as an interface between the low-power control signals from the Arduino and the higher current requirements of the motors. It ensures safe, stable, and efficient motor operation while preventing back EMF from damaging the microcontroller.

Specifications:

- Operating Voltage (Motor Supply): 2.5V – 12V
- Logic Input Voltage: 2.5V – 5V (compatible with Arduino Uno)
- Continuous Output Current per Channel: up to 800 mA
- Peak Output Current: 1.5A (short duration)
- Number of Channels: 2 (Dual H-Bridge for two motors)
- Control Logic:
 - Forward: IA = HIGH, IB = LOW
 - Reverse: IA = LOW, IB = HIGH
- Protection: Built-in suppression for back EMF

- Physical Size: Approx. 2.5×2.5 cm, lightweight and compact



Fig 8: Boat Structure

8. Boat Structure (Thermocol and Cardboard)

Description: The body of the RC boat is made using thermocol and cardboard. Thermocol provides buoyancy and stability in water, while cardboard is used for mounting and supporting electronic components.

Specifications:

- Material: Thermocol (lightweight, buoyant) + Cardboard (rigid, durable)
- Length: ~25–30 cm
- Width: ~10–15 cm
- Weight: Lightweight (< 500 g total structure)
- Waterproofing: Basic protection for electronics using plastic covering.



Fig 9: Rocker Switch

9. Rocker Switch

Description:

The rocker switch is a compact on/off switch used to control the power supply to the RC boat's Arduino and motor driver circuit. In our water pollution monitoring project, it allows the user to safely turn the system on or off. Its simple design provides a clear visual indication of the ON/OFF state and ensures reliable operation in small embedded projects.

Specifications:

- **Type:** Single Pole Single Throw (SPST), ON/OFF
- **Operating Voltage:** 12V DC
- **Current Rating:** 5A
- **Actuator:** Rocking lever
- **Mounting Type:** Panel mount
- **Terminals:** Solder or quick-connect
- **Size:** Approx. 20 × 15 mm
- **Material:** Flame-retardant plastic

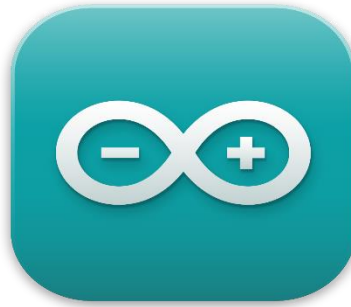


Fig 10: Arduino IDE

10. Software (Arduino IDE)

Description: The Arduino IDE is used to write, compile, and upload code to the Arduino Uno. It supports C/C++ programming and provides a simple platform for sensor reading, motor control, and Bluetooth communication.

Specifications:

- Platform: Open-source
- Programming Language: C/C++
- Compatible Boards: Arduino Uno, Mega, Nano, etc.
- Features: Serial Monitor, Debugging, and Library Support

3.8 Architecture Design

The architecture design provides a high-level view of the overall structure of the system, showing how different hardware and software components interact to achieve the project's objectives. In this project, the Arduino Uno acts as the central controller, interfacing with the turbidity sensor for data collection, the HC-05 Bluetooth module for communication, and the DC motors for boat movement. The design ensures smooth coordination between sensing, processing, communication, and control, making the system efficient and reliable.

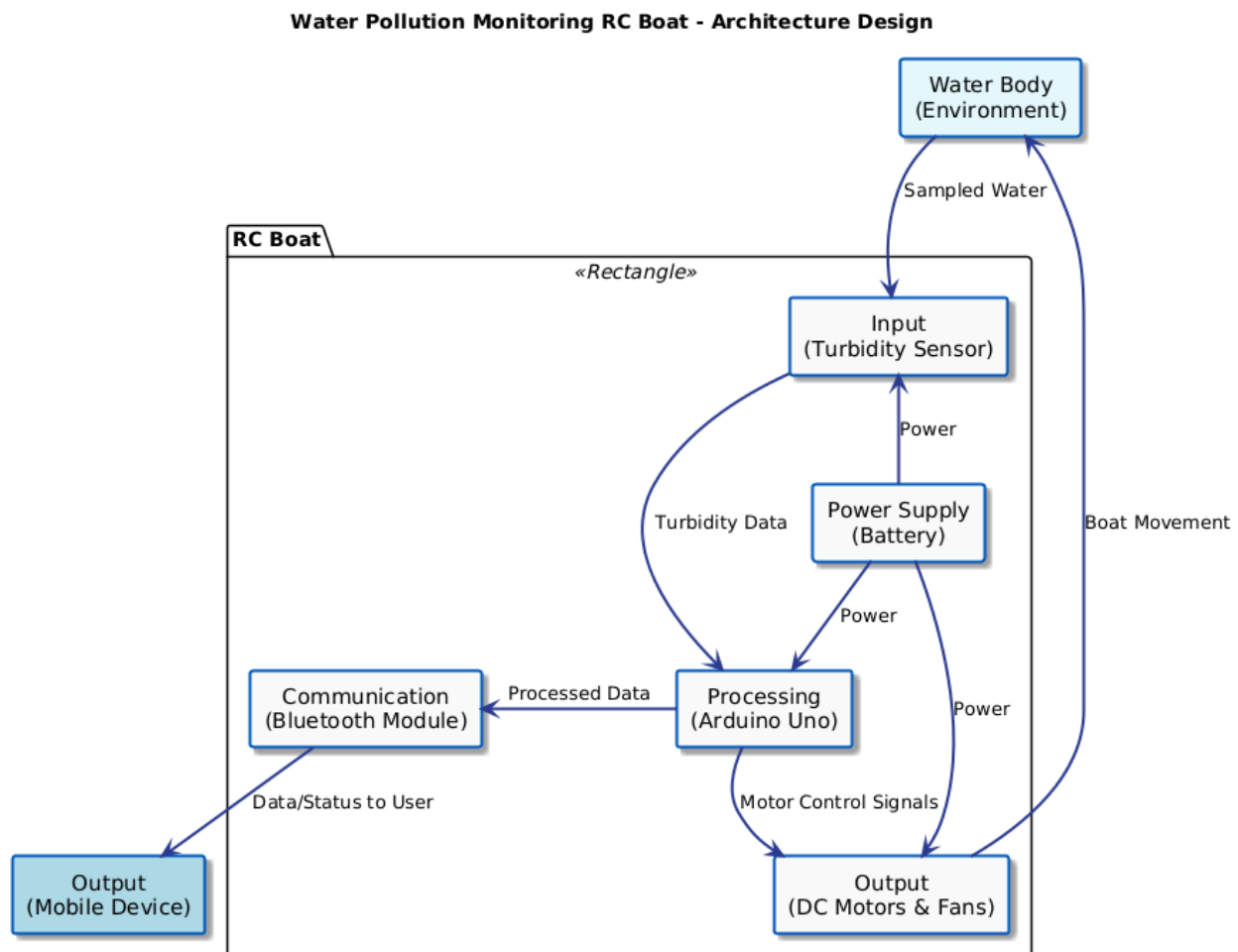


Fig 3.10: Architecture design

3.9 Block Diagram

The block diagram represents the functional structure of the project by dividing it into different blocks such as input, processing, communication, and output. It shows how the turbidity sensor collects data, the Arduino Uno processes it, the Bluetooth module transmits information to the mobile device, and the DC motors with fans enable the movement of the RC boat.

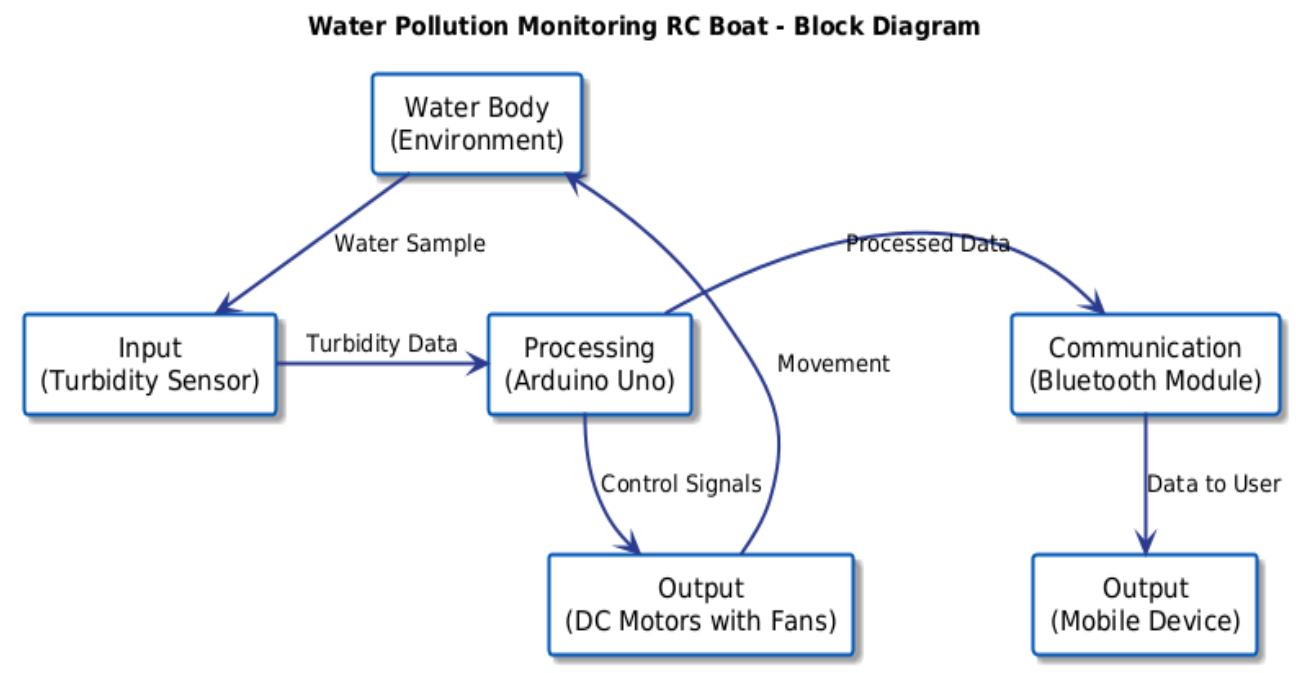


Fig 3.11: Block Diagram

4. SYSTEM PLANNING:-

4.1 GANTT CHART:

The Gantt chart is a project management tool that illustrates the timeline and progress of various activities involved in the project. It provides a clear visual representation of tasks, their duration, dependencies, and deadlines. For the water pollution monitoring RC boat project, the Gantt chart outlines stages such as research, hardware assembly, coding, integration of sensors, testing, and documentation. This helps the team to organize work efficiently, monitor progress, and ensure that all tasks are completed within the planned schedule.

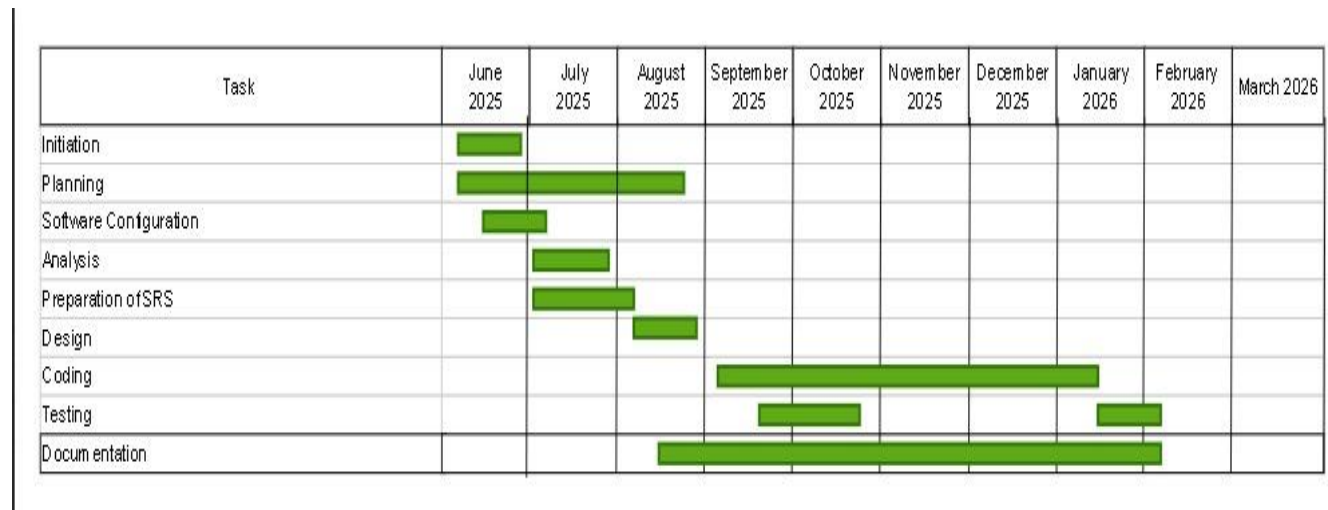
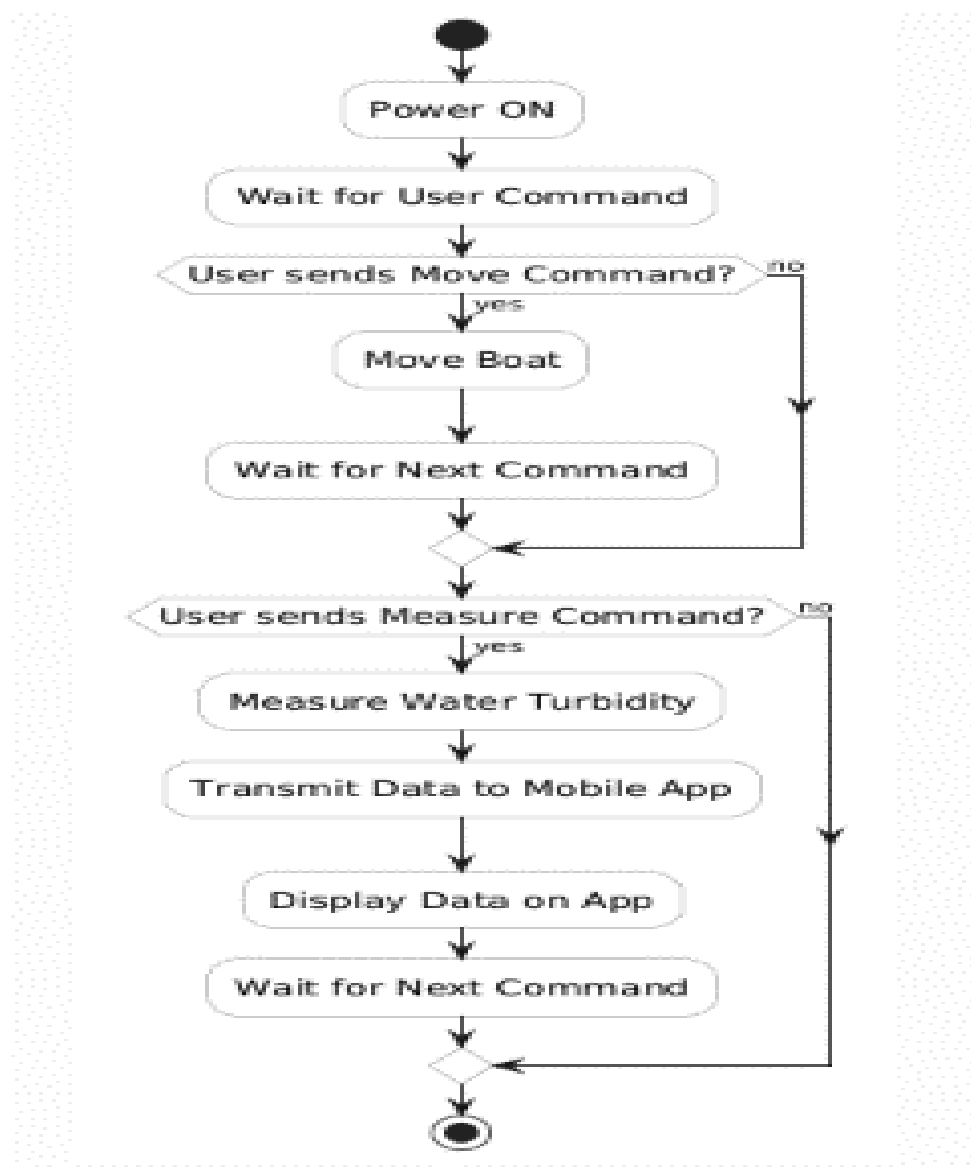


Fig 4.1: Gantt Chart

4.2 Activity Diagram:

An Activity Diagram represents the flow of activities or operations in a system, showing the sequence of actions and decision points. In the Water Pollution Monitoring Using RC Boat project, the activity diagram explains the flow starting from powering on the system, collecting turbidity sensor data, transmitting it via Bluetooth, receiving commands from the user, and controlling the boat's forward or backward movement.



5. SYSTEM IMPLEMENTATION/

MODULE WISE EXPLANATION & TESTING: -

5.1 Implementation Approaches:

The implementation of the Water Pollution Monitoring System using RC Boat was carried out in a step-by-step modular approach. Rather than developing the entire prototype at once, the work was divided into smaller stages, each focusing on a particular part of the system. This method allowed us to test and validate every module separately, ensuring that any errors could be identified early without disturbing the rest of the design. The modular process also gave flexibility to make improvements while progressing toward the final working model.

The first step of implementation involved preparing the boat structure. Since the system needed to float on water while carrying all electronic parts, materials such as thermocol and cardboard were selected for their lightweight nature, buoyancy, and cost-effectiveness. Once the boat body was ready, attention shifted toward installing the electronic modules and ensuring balanced placement of the components.

At the core of the system is the Arduino Uno microcontroller, which controls and coordinates all activities. To enable wireless operation, the HC-05 Bluetooth module was connected to the Arduino, allowing the boat to be maneuvered using a smartphone. For propulsion, two DC motors with propeller fans were added, and an L9110 motor driver was used as an interface to supply sufficient power to the motors from the batteries. This setup enabled smooth forward and backward movement of the boat.

For water quality monitoring, a turbidity sensor was installed at the front of the boat. The sensor measures suspended particles in the water and sends readings to the Arduino, which then transmits the data wirelessly to the mobile device. To ensure uninterrupted functioning during real water testing, waterproofing measures such as synthetic covers were applied around the circuit, and extra care was taken in weight distribution to keep the boat stable.

Through this modular process, the project gradually evolved from a basic floating structure into a functional prototype capable of movement, sensing, and wireless communication. The captured

photographs illustrate these stages, showcasing how the idea was transformed into a practical water pollution monitoring system.

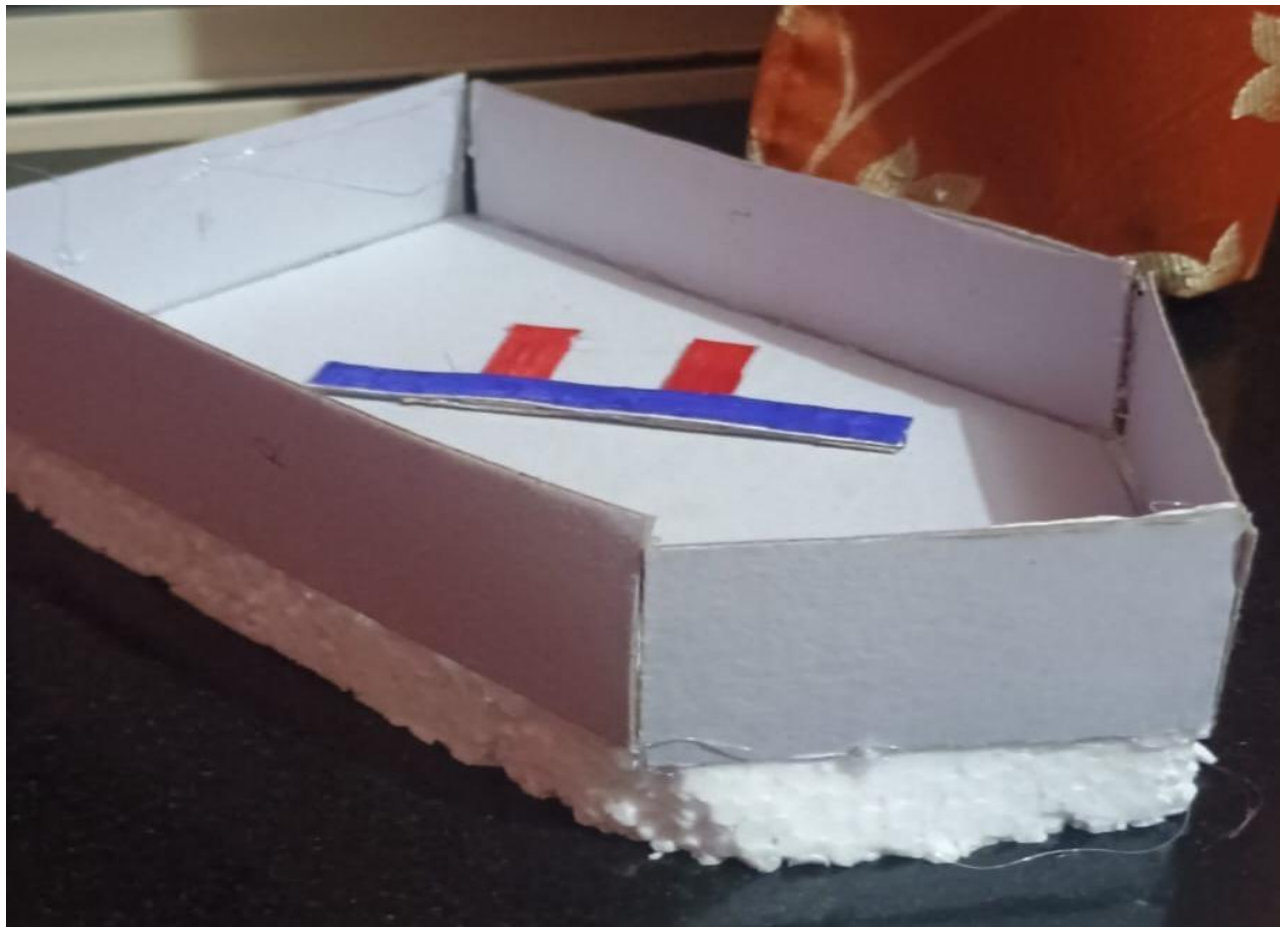


Figure 1: Structural Design of the Boat

Stage 1: Structural Design of the Boat

The first stage involved creating the basic boat structure using thermocol and cardboard. Thermocol was chosen for buoyancy to keep the boat afloat, while cardboard provided rigidity and a platform for mounting electronic components. At this stage, the boat did not include any electronics, motors, or sensors. The focus was on ensuring balance, weight distribution, and stability of the physical body of the boat in water.

This step was crucial because a weak structure would have resulted in tilting, sinking, or difficulty in carrying electronic components. Care was taken to cut and align the thermocol and cardboard properly so that it could later house the batteries, motors, and sensors.

Testing in this stage: The boat frame was placed in water to confirm buoyancy and balance. Since no electronics were added, the test was purely structural, ensuring the base could float even when additional weight would be added later. The test was successful, and the boat was ready for the next stage.

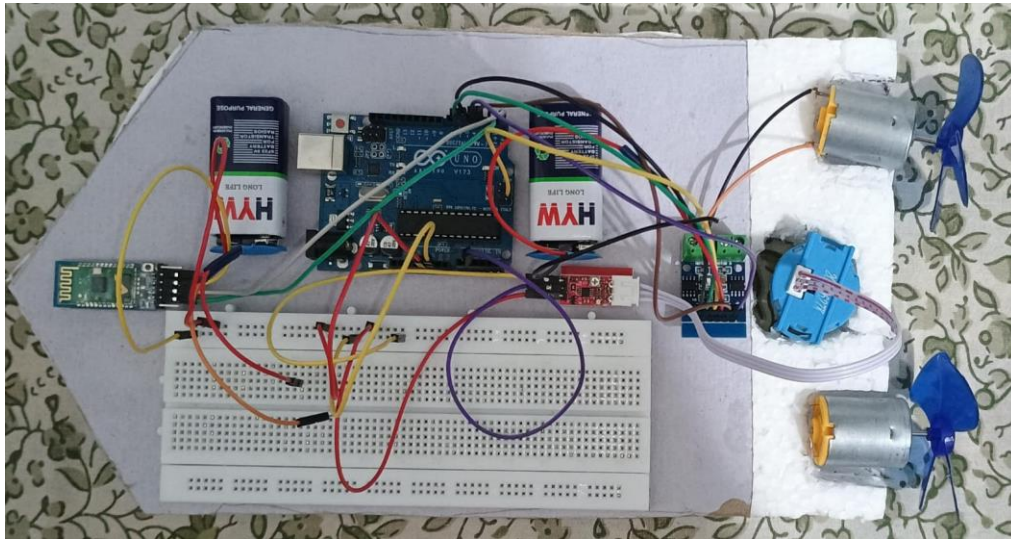


Figure 2: Placement of Components

Stage 2: Placement of Components

Once the base structure was finalized, the next step was to place the Arduino Uno, breadboard, HC-05 Bluetooth module, L9110S motor driver, DC motors with propeller fans, turbidity sensor, two 9V batteries, and jumper wires on the boat. This marked the transition from a simple structure to a functional electronic prototype.

The motors were mounted at the rear side of the boat with propeller fans attached, enabling forward and backward motion in water. The Arduino Uno was placed at the center to act as the brain of the system, while the breadboard was used for connections between modules. The Bluetooth module was positioned in a way that it could receive commands clearly from a mobile device. The turbidity sensor was connected at the front side so that it could directly measure water quality.

Testing in this stage:

- Power was supplied through the batteries, and the Arduino was programmed using the Arduino IDE.
- Individual testing was conducted to check if the Bluetooth module could pair with the mobile phone and transmit data.
- The turbidity sensor was tested in clean and muddy water to ensure variations were detected.
- The DC motors were run forward and backward through Bluetooth commands to validate navigation.

This stage confirmed that the hardware components were working as expected and that integration between modules was smooth.

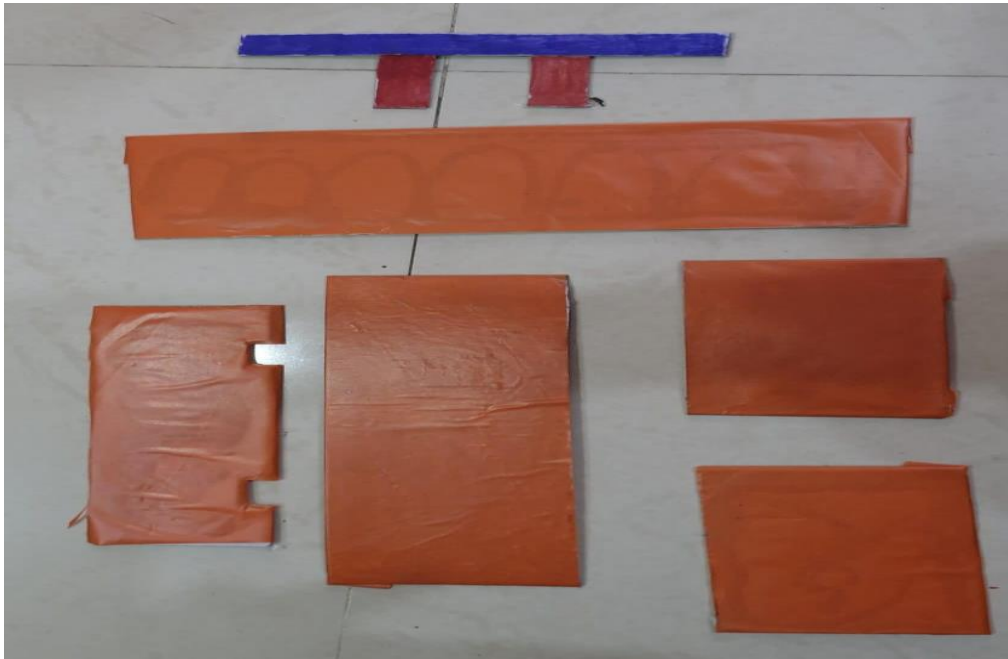


Figure 3: Water Protection and Final Assembly

Stage 3: Water Protection and Final Assembly

After placing all electronic components, the boat was further reinforced to make it suitable for real water testing. The walls of the boat were covered using synthetic waterproof sheets. These

coverings prevented water from entering the boat and damaging sensitive electronic parts such as the Arduino and Bluetooth module. The modular walls were designed and attached around the boat, enclosing the circuitry while still allowing proper wiring.

This stage finalized the implementation and made the boat durable and ready for deployment. The synthetic covering not only provided waterproofing but also added strength and aesthetic value to the boat structure.

Testing in this stage:

- The fully assembled boat was tested in water for its ability to float with the added load of components.
- Forward and backward movement was successfully achieved by controlling the motors through mobile Bluetooth commands.
- The turbidity sensor transmitted readings to the mobile device in real time.
- Continuous testing was conducted to verify battery life, stability, and waterproof performance.

The system performed reliably under test conditions, meeting the objectives of mobility, data transmission, and water quality monitoring.

Module Wise Explanation and Testing

1. Arduino Uno Module

- Role: Central controller of the system.
- Testing: Verified data reception from the turbidity sensor and commands from Bluetooth.

2. Bluetooth Communication Module (HC-05)

- Role: Wireless link between mobile device and boat.
- Testing: Successfully paired with mobile, transmitted commands, and sent turbidity values.

3. Motor Control Module (L9110S + DC Motors with Fans)

- Role: Provided movement in forward and backward directions.

- Testing: Motors rotated as per user input, propelling the boat effectively.
- 4. **Power Supply Module (Two 9V Batteries)**
 - Role: One battery powered Arduino + sensors, and the other powered the motors.
 - Testing: Stable supply ensured no interference between motors and sensors.
- 5. **Turbidity Sensor Module**
 - Role: Measured water clarity and detected suspended particles.
 - Testing: Produced accurate readings in both clean and polluted water samples.
- 6. **Structural Module (Thermocol + Cardboard + Synthetic Cover)**
 - Role: Provided buoyancy, strength, and waterproofing.
 - Testing: Boat floated smoothly without water leakage or tilting.

5.2 Coding Details and Code Efficiency:

CODE:

// --- Pin Definitions ---

#define TURBIDITY_PIN A1 // Turbidity sensor analog input

// L9110S motor driver pins for Motor A connected to Arduino pins 3 and 4

#define A_IA 3 // L9110S Motor A IA

#define A_IB 4 // L9110S Motor A IB

// L9110S motor driver pins for Motor B connected to Arduino pins 5 and 6

#define B_IA 5 // L9110S Motor B IA

#define B_IB 6 // L9110S Motor B IB

char command; // For commands received by Bluetooth

float turbidityValue = 0.0;

String status = "Unknown";

// --- Setup ---

void setup() {

Serial.begin(9600); // For HC-05 Bluetooth Transmission

pinMode(A_IA, OUTPUT);

pinMode(A_IB, OUTPUT);

pinMode(B_IA, OUTPUT);

pinMode(B_IB, OUTPUT);

}

// --- Loop ---

```

void loop() {
  // --- Read and Report Turbidity ---
  int turbidityRaw = analogRead(TURBIDITY_PIN);
  turbidityValue = turbidityRaw * 5.0 / 1023.0; // Convert to voltage

  // Status based on turbidity threshold (customize as needed)
  if (turbidityValue < 3.0) {
    status = "Good";
  } else {
    status = "Bad (High turbidity)";
  } // Send sensor info via Bluetooth
  Serial.print("Turbidity: ");
  Serial.print(turbidityValue, 2);
  Serial.print("V | Status: ");
  Serial.println(status);

  delay(1000); // Update reading every second

  // --- Bluetooth Command Handling for Motor Control ---
  if (Serial.available()) {
    command = Serial.read();

    switch (command) {
      case 'F': // Forward: Both motors forward
        digitalWrite(A_IA, HIGH);
        digitalWrite(A_IB, LOW);
        digitalWrite(B_IA, HIGH);
        digitalWrite(B_IB, LOW);
    }
  }
}

```



```
        break;
    case 'B': // Backward: Both motors backward
        digitalWrite(A_IA, LOW);
        digitalWrite(A_IB, HIGH);
        digitalWrite(B_IA, LOW);
        digitalWrite(B_IB, HIGH);
        break;
    case 'L': // Left Turn: Only right motor forward
        digitalWrite(A_IA, LOW);
        digitalWrite(A_IB, LOW);
        digitalWrite(B_IA, HIGH);
        digitalWrite(B_IB, LOW);
        break;
    case 'R': // Right Turn: Only left motor forward
        digitalWrite(A_IA, HIGH);
        digitalWrite(A_IB, LOW);
        digitalWrite(B_IA, LOW);
        digitalWrite(B_IB, LOW);
        break;
    case 'S': // Stop: Both motors off
        digitalWrite(A_IA, LOW);
        digitalWrite(A_IB, LOW);
        digitalWrite(B_IA, LOW);
        digitalWrite(B_IB, LOW);
        break;
    }
}
}
```

Coding Details and Code Efficiency

The coding of the Water Pollution Monitoring RC Boat project integrates both movement control and water quality monitoring in a single program. The code has been written in Arduino C/C++ and uploaded onto an Arduino Uno, which acts as the central processing unit. Each line of code has been designed with clarity and modularity in mind, ensuring that it can be easily understood, modified, and extended if more sensors or functionalities are added in the future.

Code Structure and Flow

1. **Pin Definitions:** At the beginning of the program, pin assignments are made to ensure organized hardware control. Four digital pins (3, 4, 5, 6) are allocated to the L9110S motor driver module, which drives two DC motors for propulsion. One analog pin (A1) is assigned to the turbidity sensor, which provides continuous voltage readings proportional to the water quality.
2. **Global Variables:** A few variables are declared to maintain simplicity. The `command` variable stores incoming Bluetooth commands (F, B, L, R, S). The `turbidityValue` variable holds the converted sensor reading, and the `status` string stores the qualitative water condition (Good/Bad).
3. **Setup Function:** The `setup()` function initializes serial communication at 9600 baud, allowing smooth data exchange with the HC-05 Bluetooth module. The motor control pins are set as outputs, ensuring that the Arduino can drive the motors as soon as valid commands are received.
4. **Loop Function:** The `loop()` function contains two main tasks:
 - **Turbidity Measurement:** The turbidity sensor's analog output is read using `analogRead()`. This raw value is converted into a voltage (0–5V). A threshold-based decision is then applied to classify the water condition as either “Good” (low turbidity) or “Bad” (high turbidity). The values and status are transmitted via Bluetooth, enabling the user to view real-time results on a paired mobile device.
 - **Motor Control:** The program continuously checks for incoming Bluetooth commands. Depending on the character received:
 - **F (Forward):** Both motors rotate forward.

- **B (Backward):** Both motors rotate in reverse.
- **L (Left):** Right motor runs while left motor is stopped.
- **R (Right):** Left motor runs while right motor is stopped.
- **S (Stop):** Both motors are turned off.

A `switch-case` statement is used for command processing, which improves readability and avoids long chains of `if-else` conditions.

Code Efficiency

The program is lightweight and well-optimized for the limited resources of the Arduino Uno. Instead of storing or processing large data sets, it performs direct sensor readings and immediately transmits results to the user, ensuring minimal memory usage.

1. **Memory and Speed Optimization:** Only a few essential variables are declared (`command`, `turbidityValue`, `status`), which reduces the RAM footprint. Sensor readings and motor control operations are executed independently, ensuring that motor responses do not lag due to sensor calculations.
2. **Real-Time Monitoring:** By transmitting sensor readings every second, the system provides continuous water quality feedback without overwhelming the Bluetooth communication channel.
3. **Separation of Modules:** The navigation logic (motor control) is cleanly separated from the monitoring logic (sensor readings). This modularity improves readability, simplifies debugging, and makes the system highly extendable. Adding more sensors like **pH**, **temperature**, or **dissolved oxygen** would only require minor code additions, without affecting the motor control logic.
4. **Reliability and Simplicity:** The use of direct voltage thresholds ensures that turbidity classification is quick and does not require complex computation. The use of `switch-case` for motor commands makes the system highly responsive and user-friendly.

5.3 Testing Approach:

Testing is an essential phase of any project development lifecycle, as it ensures that the system works as expected, fulfills its requirements, and performs reliably under real-world conditions. In the case of the Water Pollution Monitoring RC Boat project, testing plays a crucial role because the system integrates hardware, software, sensors, and communication modules that must work together smoothly. The primary aim of testing is to verify the correctness of the design, validate the functionality of each module, and ensure the overall system operates efficiently and safely.

Different testing approaches are adopted in this project to thoroughly examine each part of the system. These approaches cover both hardware and software, as well as the interaction between them. The main testing approaches used include unit testing, integration testing, system testing, hardware testing, communication testing, and field testing. Each type of testing is described below.

1. Unit Testing

Unit testing was first performed on each module separately to confirm proper functioning:

- **Turbidity Sensor:** The sensor was connected to the analog pin of the Arduino Uno, and readings were observed through the Serial Monitor for both clean and contaminated water samples. This verified that the sensor gave measurable outputs corresponding to water clarity.
- **Motor Driver (L9110S) and DC Motors:** The L9110S motor driver, which acts as an interface between Arduino and the two DC motors, was tested independently. Digital pins from the Arduino were programmed to send HIGH and LOW signals to the input pins of the driver. The driver amplified these control signals and powered the motors accordingly. The motors were tested for forward rotation, reverse rotation, and stopping to ensure they responded correctly.
- **Bluetooth Module (HC-05):** Tested separately by sending characters from a mobile device and verifying correct reception in the Arduino Serial Monitor. This confirmed that the module was capable of reliable wireless communication.

2. Integration Testing

Once individual modules were verified, they were combined to check whether they functioned smoothly together. For example:

- The Arduino simultaneously controlled the L9110S motor driver for boat navigation while also reading turbidity sensor values.
- Bluetooth commands such as `F` (forward), `B` (backward), and `S` (stop) were received, processed by the Arduino, and sent as control signals to the L9110S driver. The motors responded accurately while the sensor continued to collect data without interruption.
- Testing ensured there was no conflict in handling both motor control and sensor communication at the same time.

3. System Testing

In this phase, the complete boat setup was tested in real operating conditions. The RC boat was placed in water, and the overall system was evaluated for:

- Forward and backward movement controlled by Bluetooth through the motor driver.
- Stability of the boat when both motors were active.
- Real-time transmission of turbidity sensor readings while the boat was moving. The system testing confirmed that the objectives of navigation and water quality monitoring were achieved together.

4. Hardware Testing

Hardware reliability was thoroughly tested because the system depends on proper interaction between multiple components:

- The **Arduino Uno** was checked for stable operation under continuous execution.
- The **L9110S motor driver** was tested for its ability to provide sufficient current to both DC motors without overheating.

- The **DC motors with propeller fans** were tested for endurance and response time. Repeated forward and backward movements were performed to verify durability.
- The **9V batteries** were tested for supplying consistent power to both logic circuits (Arduino + Bluetooth + sensor) and high-current components (motors).
- The **boat body** (thermocol + cardboard) was tested for buoyancy and stability under the load of electronics and motors.

5. Communication Testing

The communication between the HC-05 Bluetooth module and the Arduino was tested in terms of response time, accuracy, and range. Commands were sent from a mobile device, and the time taken for the boat to respond was recorded. Tests confirmed reliable operation within a 10-meter range, with stable transmission of both control commands and turbidity readings.

Importance of Testing Approaches

By following these testing approaches, potential issues were identified and corrected at every stage. The L9110S motor driver testing ensured safe and efficient motor control, while integration and field testing confirmed that all modules worked together harmoniously. Ultimately, the thorough testing process validated that the project is not only functional but also reliable, cost-effective, and applicable for real-world water pollution monitoring.

5.4 Test Cases:

Introduction

Testing is an essential stage in any embedded system or IoT project because it validates the functionality, reliability, and efficiency of the system. For the Water Pollution Monitoring RC Boat, testing ensures that all hardware and software components work together seamlessly. This project involves multiple modules, including the Arduino Uno microcontroller, HC-05 Bluetooth module, turbidity sensor, L9110S motor driver with DC motors, and the power supply system.

The purpose of these test cases is to systematically evaluate each module individually (unit testing), test interactions between modules (integration testing), and verify the complete system in real operating conditions (system testing). Each test case specifies the input, expected output, actual output, and whether the system passed or failed the test. By carrying out these tests, potential errors or inefficiencies are detected early, ensuring that the RC boat functions correctly for monitoring water pollution.

Test Case Scenarios

The testing of the RC Boat is divided into three categories:

1. **Unit Testing:** To check individual components such as turbidity sensor calibration, motor rotation, and Bluetooth pairing.
2. **Integration Testing:** To verify that modules communicate effectively with one another, e.g., Arduino receiving commands via Bluetooth and driving motors accordingly.
3. **System Testing:** To ensure that the entire system, when deployed in real water bodies, behaves as expected without performance breakdowns.

Test Case ID	Description	Input / Condition	Expected Output	Actual Output	Result
TC-01	Turbidity sensor reading in clean water	Place sensor in clear water	Low turbidity value (~150–200)	Display shows ~180	Pass
TC-02	Turbidity sensor reading in polluted water	Place sensor in muddy water	High turbidity value (~700–800)	Display shows ~740	Pass
TC-03	Sensor reading in semi-polluted water	Place sensor in slightly turbid water	Medium value (~400–500)	Display shows ~460	Pass
TC-04	Forward movement	Send command F via Bluetooth	Boat moves forward	Boat moves forward smoothly	Pass
TC-05	Backward movement	Send command B via Bluetooth	Boat moves backward	Boat moves backward	Pass
TC-06	Bluetooth connectivity	Pair HC-05 with mobile	Successful pairing and command transmission	Connection stable, commands received	Pass
TC-07	Data transmission	Sensor sends turbidity values via Bluetooth	Correct values displayed on mobile	Values displayed correctly	Pass
TC-08	Power stability	Operate system continuously for 20 minutes	Stable operation, no resets	System stable for full duration	Pass
TC-09	Real-time system operation	Deploy boat in water with all modules active	Boat moves as commanded and sends turbidity data	Successful movement and data reporting	Pass

Explanation of Results

The test cases confirm that the project operates reliably in all intended scenarios. The turbidity sensor provides accurate readings across clean, polluted, and moderately turbid water samples. The DC motors controlled via the L9110S motor driver respond promptly to forward and backward commands from the Bluetooth module. This ensures precise navigation in the water without manual intervention.

The HC-05 Bluetooth module has demonstrated stable wireless communication, allowing both command reception and data transmission to a mobile device. System-level testing in a water tank or pond confirmed that the boat can perform real-time water monitoring while moving, and the power supply is sufficient to sustain operation for practical durations.

Through these tests, the functionality, reliability, and efficiency of the RC boat system have been validated. The results demonstrate that the project meets its design objectives of providing a safe, mobile, and cost-effective solution for monitoring water pollution.

5.5 Modifications and Improvements:

The Water Pollution Monitoring RC Boat project is designed as a functional prototype for assessing water quality using a turbidity sensor and remote-controlled navigation. While the current system meets the basic objectives of collecting water quality data and moving forward and backward using Bluetooth commands, there is significant scope for modifications and improvements to enhance the system's efficiency, reliability, and overall applicability. The purpose of this section is to explore potential enhancements and refinements that can be incorporated into the project, either for academic purposes or practical deployment in real-world water bodies.

1. Hardware Improvements

Boat Structure and Stability: Currently, the boat is constructed using thermocol and cardboard, which provide a lightweight and inexpensive platform. However, these materials are sensitive to water exposure and may degrade over time. One improvement could be the use of waterproof and lightweight materials such as foam sheets, plastic, or fiberglass, which would enhance durability and allow for longer operational periods. Additionally, the boat's hull design can be modified to improve stability and maneuverability, reducing the risk of tipping in turbulent water.

Motor and Propulsion System: The existing design uses two DC motors with fans to enable forward and backward motion. Future modifications could include more powerful motors or propellers for increased speed and efficiency, especially in larger water bodies. Adding motor control for variable speed would allow the boat to adjust its movement according to water resistance or specific sampling requirements, enhancing control and energy efficiency.

Power Supply: Currently, the system relies on two 9V batteries to power both the Arduino and the motors. To extend operational time, the project could incorporate rechargeable Li-ion or Li-Po battery packs, which provide higher capacity and longer life. Additionally, adding a voltage monitoring circuit would prevent sudden shutdowns and ensure the electronics are protected from low-voltage conditions.

Sensor Enhancements: While the turbidity sensor effectively measures water clarity, additional sensors could improve water quality assessment. For example, integrating pH, dissolved oxygen, or temperature sensors would allow for a more comprehensive analysis. These additional measurements could help detect chemical contamination, oxygen depletion, and environmental changes in aquatic ecosystems.

2. Software Improvements

Bluetooth Communication: The HC-05 module currently handles both navigation and sensor data transmission. A potential improvement is the development of a dedicated mobile application with a user-friendly interface that can display real-time turbidity readings, log historical data, and allow remote control of the boat. Additionally, enhancing the communication protocol could improve data reliability, prevent signal loss, and extend the operational range of the system.

Automation and Data Logging: Currently, the Arduino receives commands manually via Bluetooth. Improvements could include automated navigation patterns, where the boat follows a predefined route while continuously collecting data. Integration of data logging on an SD card or microcontroller memory would allow long-term storage for later analysis, reducing dependency on continuous Bluetooth connectivity.

Code Optimization: The current Arduino code could be refined for efficiency, such as reducing latency in sensor readings, implementing interrupt-driven motor control, and ensuring power-efficient loops. These improvements would enhance responsiveness, reduce energy consumption, and improve system reliability during extended operation.

3. Operational and Safety Improvements

Waterproofing and Protection: All electronic components, including the Arduino, motor driver, and batteries, could be enclosed in waterproof casing to prevent accidental short circuits or water damage. Protective design modifications would make the boat safer for operation in natural water bodies like ponds, rivers, or reservoirs.

Modular Design: Implementing a modular hardware layout allows components to be easily replaced or upgraded. For example, the motor or sensor module could be detached and replaced with improved versions without redesigning the entire boat. This modularity simplifies maintenance and enables future upgrades.

Scalability and Multi-Boat Systems: Future improvements could involve multiple RC boats operating simultaneously in larger water bodies. Coordination between boats could enable comprehensive water monitoring coverage and faster data collection, providing a more effective environmental management solution.

4. Future Prospects

By implementing these modifications and improvements, the RC boat can evolve from a simple academic prototype into a robust, practical tool for environmental monitoring. The combination of enhanced hardware durability, improved sensor suite, automated navigation, and advanced software interfaces will make the system more efficient, user-friendly, and adaptable to diverse water quality monitoring scenarios. Additionally, the project could serve as a foundation for further research, such as integrating IoT cloud platforms for remote monitoring and real-time alerts, or developing AI algorithms to analyze water quality trends and detect pollution sources.

6. Cost and Benefit Analysis:

Cost Analysis:

Here is the detailed cost breakdown of the components we used in our project:

Component / Resource	Quantity	Cost (INR)	Remarks
Arduino Uno	1	450	Microcontroller for controlling sensors and motors
L9110 Motor Driver	1	50	Drives two DC motors efficiently
DC Motors with fans	2	60	Provides propulsion for the RC boat
HC-05 Bluetooth Module	1	250	Wireless communication with mobile app
Turbidity Sensor	1	500	Measures water pollution levels
9V Battery	2	60	Power supply for Arduino and motors
Rocker Switch	1	20	Manual power ON/OFF control
Breadboard & Jumper Wires	1 set	150	For circuit connections
Boat Chassis / RC Boat Body(Thermocol, Cardboard)	1	130	Platform to mount all components
Miscellaneous	–	250	Small supporting hardware
Documentation & Reporting	–	500	
Total Project Cost			₹2,400

Bottom-Up Cost Estimation

Bottom-Up Cost Estimation is a widely used method for determining the total cost of a project by estimating the cost of each individual component or task and then adding them together. This approach provides a detailed and accurate overview of the resources required, helping ensure that no component or minor expense is overlooked. For our Water Pollution Monitoring RC Boat Project, Bottom-Up Estimation allows us to account for all hardware components, sensors, electronic modules, and miscellaneous items, resulting in a precise calculation of the total project cost.

In this method, each component's cost is individually assessed based on current market prices. The Arduino Uno acts as the central controller of the system, managing the motor operations, sensor readings, and communication with the mobile device. Its cost forms a significant portion of the total project budget. The L9110 motor driver and DC motors are essential for the movement of the boat, providing the required forward and backward propulsion as directed by the Arduino.

The HC-05 Bluetooth module enables wireless communication between the boat and a mobile application, allowing remote control of the system. The turbidity sensor is used to measure water quality by detecting the amount of suspended particles, which is an indicator of pollution. 9V batteries are used to supply power to the Arduino, motor driver, and motors, while a rocker switch provides a simple and reliable way to turn the system on and off manually.

Additional supporting components, such as breadboards, jumper wires, LEDs, resistors, and the RC boat chassis, complete the system. These components are small in cost individually, but their inclusion is necessary for assembling and connecting the entire hardware system properly.

By summing the cost of all these components, the **total estimated cost of the project is ₹2,400**. This estimation provides clarity on where the majority of the budget is allocated and highlights areas where cost optimization could be considered, such as selecting alternative suppliers or using lower-cost components without compromising system performance.

The Bottom-Up Cost Estimation method is particularly suitable for small-scale embedded systems projects like ours. It allows students and developers to carefully plan the budget, ensures

transparency in financial management, and prevents unforeseen expenses during implementation. Furthermore, it helps in identifying the critical components that contribute most to the overall cost and encourages effective resource management.

In conclusion, using Bottom-Up Cost Estimation for the Water Pollution Monitoring RC Boat Project ensures that the project is financially planned, components are adequately accounted for, and the total expenditure remains within the estimated budget. This method not only aids in procurement planning but also provides a clear framework for analyzing cost efficiency and resource utilization.

Benefit Analysis

- **Educational Benefits:**

- Hands-on experience with Arduino programming and embedded systems.
- Understanding of DC motor control, H-bridge drivers, and sensor interfacing.
- Practical knowledge of IoT applications and Bluetooth communication.

- **Environmental Benefits:**

- Enables real-time monitoring of water quality in local water bodies.
- Helps identify pollution levels, aiding in environmental protection initiatives.

- **Technical and Functional Benefits:**

- Compact and portable system for field measurements.
- Low-cost alternative to commercial water monitoring devices.
- Dual functionality: remote-controlled boat movement + turbidity measurement.

- **Project Benefits for Students:**

- Improves problem-solving and project design skills.
- Provides a complete mini-project suitable for exhibitions or competitions.
- Encourages innovative applications in environmental monitoring using embedded systems.

7. Results and Discussion:

7.1 Test Reports:

Testing forms a crucial part of the project development process, as it ensures that both hardware and software components perform as intended under real operating conditions. For the Water Pollution Monitoring Using RC Boat project, testing was carried out at different stages—module-level testing, integration testing, and real-time system testing. The reports presented here describe how each part of the system was evaluated, what outcomes were expected, and how the observed results compared with those expectations. The emphasis was on validating the turbidity sensor readings, Bluetooth communication efficiency, motor driver and propulsion performance, as well as the overall functionality of the boat in natural water environments.

1. Turbidity Sensor Testing

The turbidity sensor is a vital component of the project since it directly measures the cloudiness of water, which is an indicator of pollution. During testing, the sensor was immersed in different types of water samples: clear drinking water, mildly contaminated water (with a small amount of soil or impurities), and heavily turbid water (containing mud or waste particles). The analog values generated by the sensor were read by the Arduino Uno and transmitted to a mobile device via Bluetooth.

- **Expected Result:** Clear water should provide low turbidity values, while polluted or muddy water should show higher readings.
- **Observed Result:** The sensor behaved as expected, producing distinct values across the three samples. For clear water, the readings were minimal; mildly contaminated water showed medium values; and heavily turbid water generated significantly higher readings.

This confirmed that the turbidity sensor was reliable in differentiating water quality levels. The results also proved the system's ability to measure water pollution in real time.

2. Bluetooth Communication Testing

The **HC-05 Bluetooth module** plays a dual role in the system—it transmits turbidity sensor data to the user's mobile phone and receives navigation commands to control the boat. Testing was performed by pairing the module with a smartphone and using a Bluetooth terminal application to interact with the Arduino Uno. Commands for forward and backward movement were sent, while at the same time turbidity readings were displayed on the mobile screen.

- **Expected Result:** The Bluetooth module should maintain a stable connection, transmit data quickly, and respond to control commands without delay.
- **Observed Result:** The module maintained a continuous connection with minimal latency. Commands for boat navigation were executed almost instantly, and turbidity readings were displayed on the mobile device in real time.

This verified that Bluetooth communication was efficient, stable, and sufficient for both control and monitoring purposes.

3. DC Motors and L9110 Motor Driver Testing

The propulsion system of the boat was tested to ensure that the DC motors, controlled through the L9110 motor driver, operated according to the user's commands. The Arduino Uno was programmed to change motor directions depending on Bluetooth inputs. One motor was connected to each channel of the L9110, with both powered by a dedicated 9V battery to ensure stable performance.

- **Expected Result:** The motors should rotate in the correct direction and provide sufficient thrust to move the boat forward and backward.
- **Observed Result:** Both motors performed as expected. The boat moved smoothly forward when the command was given, and similarly reversed direction when instructed to move backward. The L9110 effectively amplified Arduino control signals into motor-driving currents, while also protecting the circuit from back EMF. This confirmed that the propulsion system was functioning reliably and was well integrated with the rest of the setup.

4. Integrated System Testing

The final stage involved testing the complete system in real-world conditions. The RC boat, with all its components installed, was floated on a pond. Commands were sent via Bluetooth to move the boat forward and backward while collecting turbidity data from different points in the water body.

- **Expected Result:** The RC boat should respond to movement commands smoothly while transmitting turbidity data continuously.
- **Observed Result:** The boat successfully navigated to different parts of the pond, and real-time turbidity readings were received without interruption. The system demonstrated efficiency in both navigation and monitoring. Even in slightly wavy conditions, the boat remained stable and data transmission was unaffected.

This stage confirmed the practical viability of the project as a working prototype for water pollution monitoring.

Discussion of Results

The test reports highlight that all modules of the project performed satisfactorily, meeting the objectives set during the design phase. The turbidity sensor effectively detected water clarity differences, proving its usefulness in monitoring pollution levels. The Bluetooth module, despite being a low-cost option, was capable of handling simultaneous data transmission and navigation control. The DC motors, powered through the L9110 driver, ensured stable propulsion of the boat. Finally, integrated system testing validated the overall functionality of the RC boat in real water environments, making the project both reliable and cost-effective.

Test ID	Test Component	Expected Result	Observed Result	Status	Remarks
TR-01	Turbidity Sensor	Clear water gives low readings; turbid water gives high readings.	Sensor values increased with higher turbidity, matched expectations.	Pass	Sensor works reliably.
TR-02	Bluetooth Communication	Should transmit turbidity data and receive forward/backward commands without delay.	Commands executed instantly, data received continuously with minimal delay.	Pass	Stable connection throughout testing.
TR-03	DC Motors + L9110 Driver	Motors should rotate forward/backward as per commands and provide stable propulsion.	Motors worked as expected, boat moved smoothly in both directions.	Pass	Needs regular battery check for stability.
TR-04	Integrated Boat Testing	Boat should navigate and send data simultaneously in real water.	Boat successfully navigated pond, sent turbidity readings in real time without errors.	Pass	Stable even in slightly wavy conditions.

7.2 User Documentation:

Introduction

This User Documentation provides a step-by-step guide to understanding, setting up, and using the Water Pollution Monitoring RC Boat system. The project is designed to help users monitor the turbidity level of water bodies in real-time using a turbidity sensor, while also being able to control the boat's movement (forward and backward) through Bluetooth commands. This documentation ensures that even non-technical users can operate the system efficiently and safely.

System Requirements

Before operating the system, ensure that the following components and requirements are met:

- **Hardware Components:**
 - Arduino Uno microcontroller
 - HC-05 Bluetooth module
 - Turbidity sensor
 - L9110S motor driver
 - Two DC motors with propeller fans
 - Breadboard and jumper wires
 - Two 9V batteries
 - Boat body made from thermocol and cardboard
- **Software Requirements:**
 - Arduino IDE (for uploading code to Arduino)
 - Mobile device with Bluetooth capability (for sending commands and receiving turbidity data)

Setup Instructions

1. **Assembly of Hardware:**
 - Place the Arduino Uno and breadboard securely on the boat structure.
 - Connect the turbidity sensor to the Arduino as per the circuit diagram.

- Attach the HC-05 Bluetooth module for wireless communication.
 - Connect the L9110S motor driver to both DC motors and to the Arduino.
 - Ensure that the 9V batteries are properly connected to power the system.
2. **Software Setup:**
- Upload the provided Arduino program to the Arduino Uno board using Arduino IDE.
 - Ensure the correct COM port and board type are selected in the software.
3. **Bluetooth Pairing:**
- Switch on the boat's power supply.
 - Open Bluetooth settings on your mobile device and pair it with the HC-05 module.
 - Use a mobile application or Bluetooth terminal to send movement commands and receive sensor data.

Operating Instructions

- **Boat Movement:**
 - Send the **Forward command** → Both DC motors rotate in forward direction, moving the boat ahead.
 - Send the **Backward command** → Both DC motors rotate in reverse, moving the boat backward.
- **Water Quality Monitoring:**
 - Place the boat in the desired water body.
 - The turbidity sensor will detect the clarity or cloudiness of the water.
 - The readings are transmitted to your mobile device in real-time via Bluetooth.

Safety and Maintenance Guidelines

- Ensure batteries are fully charged before operation to avoid power failure during testing.
- Do not submerge the Arduino, Bluetooth module, or wiring in water. Only the turbidity sensor's probe should come in contact with water.
- Use the boat in calm or controlled water conditions to prevent instability.
- Disconnect batteries after use to prevent unnecessary discharge.

- Regularly check motor connections and wires for wear or loose contacts.

Troubleshooting

- **Boat not moving:** Check motor driver connections and ensure batteries are properly connected.
- **No Bluetooth connection:** Confirm that the HC-05 is powered and visible in your mobile's Bluetooth settings.
- **No sensor readings:** Verify wiring of turbidity sensor and check code uploaded on Arduino.
- **Weak performance:** Replace batteries or check if motors are obstructed.

8. Conclusion:

8.1 Significance of the System:

The Water Pollution Monitoring RC Boat system holds significant value in addressing one of the most critical environmental challenges of modern times—water pollution. Water is a vital resource for human life, agriculture, industry, and aquatic ecosystems, yet it is increasingly threatened by pollutants such as chemicals, industrial waste, and suspended particles. Traditional methods of monitoring water quality require manual sampling and laboratory testing, which are not only time-consuming but also pose health risks to personnel working in contaminated areas. The system developed in this project overcomes these limitations by providing a low-cost, portable, and efficient solution for real-time turbidity monitoring in various water bodies.

One of the main contributions of this system is the use of a remotely controlled boat as a mobile platform. Unlike static monitoring stations, the RC boat can access multiple points in a water body, enabling broader coverage and more accurate detection of pollution levels. By integrating the turbidity sensor, the system provides direct information about the clarity and quality of water. Turbidity is an important indicator of pollution, as high turbidity levels suggest the presence of harmful suspended particles, which can negatively affect aquatic organisms and human health. Therefore, continuous monitoring through this system can help in early detection of pollution hotspots, allowing authorities and communities to take timely corrective actions.

Another significant aspect of this project is its ability to combine mobility and monitoring through simple, user-friendly technology. The HC-05 Bluetooth module allows the boat to be navigated forward and backward while simultaneously transmitting turbidity data to a mobile device. This dual functionality not only reduces the complexity of the system but also makes it more practical for field use. The design emphasizes safety by minimizing human contact with polluted water and ensuring that data can be collected from hazardous areas without risk.

8.2 Conclusion:

The Water Pollution Monitoring RC Boat project successfully demonstrates how embedded systems, wireless communication, and sensor technology can be combined to create an effective, low-cost, and portable solution for environmental monitoring. By integrating a turbidity sensor with a remotely controlled boat, the system provides real-time information about water quality without requiring manual sample collection or exposing humans to unsafe and polluted areas. This innovation makes the monitoring process more efficient, safe, and accessible compared to conventional laboratory-based methods.

The use of the Arduino Uno as the central microcontroller ensures reliable data processing, while the HC-05 Bluetooth module plays a dual role by enabling both boat navigation and data transmission to a mobile device. This simplifies the system architecture while maintaining high functionality. The L9110S motor driver and dual DC motors provide smooth and controlled forward and backward movements, making the boat highly maneuverable in different water conditions. The lightweight structure of the boat, made from thermocol and cardboard, reduces cost while maintaining adequate stability for sensor-based measurements.

From an environmental perspective, this project holds great importance as turbidity is a key indicator of water pollution. High turbidity levels suggest the presence of harmful suspended particles, which can damage aquatic life, contaminate drinking water, and disrupt ecosystems. By providing real-time turbidity data, the system helps in identifying pollution hotspots quickly, enabling authorities and communities to take preventive or corrective measures. The project thus contributes to raising awareness about water quality issues and supports sustainable management of this critical resource.

In addition, the project highlights the educational and research value of combining hardware and software to address real-world problems. It serves as a foundation for future improvements, such as incorporating GPS for mapping, adding more sensors for advanced water analysis, or integrating cloud-based storage for long-term monitoring and analysis. These extensions can make the system even more versatile and impactful in professional environmental management.

8.3 Limitations of the System:

The water pollution monitoring RC boat is an innovative and cost-effective solution for studying water quality in small water bodies; however, it has certain limitations that should be considered when evaluating its performance and applicability. One of the primary limitations is the battery life. The system relies on 9V batteries to power the Arduino, motor driver, and DC motors. While suitable for short-duration field measurements, extended usage requires frequent battery replacement or recharging, which can be inconvenient during prolonged monitoring sessions.

Another limitation is the range of Bluetooth communication. The HC-05 Bluetooth module used for remote control generally operates within a range of approximately 10 meters under ideal conditions. This restricts the distance at which the boat can be reliably operated and limits its application in larger water bodies or areas with obstacles that may block the signal.

The system also faces environmental constraints. It is designed for calm water surfaces, such as ponds or small lakes. Strong water currents, waves, or adverse weather conditions can affect the stability, movement, and navigation of the boat. This makes the system less suitable for open rivers or turbulent waters.

Sensor accuracy is another area of concern. The turbidity sensor provides approximate measurements of water clarity, which are sufficient for general pollution monitoring but may not be precise enough for scientific research requiring highly accurate data. Calibration may improve accuracy, but it still cannot match professional water quality monitoring equipment.

The load-handling capacity of the system is limited due to the small DC motors and L9110 motor driver. Adding extra weight or additional sensors could reduce the boat's speed, stability, and overall performance, restricting the system's versatility for multiple tasks simultaneously.

Furthermore, the system relies heavily on manual operation. The boat's movement is controlled remotely, and the user must monitor and interpret sensor readings in real time. There is no autonomous navigation or automated data processing, which limits efficiency and requires constant attention during operation.

9. Future Work:

The water pollution monitoring RC boat project provides a solid foundation for environmental monitoring, but there is significant scope for improvement and expansion in future iterations. One of the main areas of enhancement is autonomous navigation. By integrating GPS modules and path-planning algorithms, the boat can be programmed to follow predefined routes in a water body without manual intervention. This would allow for systematic water quality monitoring over larger areas and reduce the dependency on human operation.

Another potential improvement is enhanced sensor integration. While the current system uses a turbidity sensor to assess water pollution, future versions could incorporate additional sensors such as pH sensors, dissolved oxygen sensors, temperature sensors, or chemical contaminant detectors. This would provide a more comprehensive analysis of water quality, enabling better decision-making for environmental protection.

Wireless communication can also be upgraded. The current HC-05 Bluetooth module limits the operational range to about 10 meters. Future work can involve implementing Wi-Fi, LoRa, or GSM-based communication to allow for long-range monitoring and real-time data transmission to cloud platforms. This would make the system suitable for remote water bodies and provide continuous monitoring capabilities.

The power management system can be improved to increase operational time. Using rechargeable lithium-ion batteries or solar panels can provide longer usage durations and make the system more sustainable. Incorporating a battery management system (BMS) could also help in monitoring battery health and avoiding unexpected shutdowns during operation.

Another avenue for future development is data logging and analysis. Currently, the turbidity readings are sent to a mobile app in real time. Adding onboard storage or cloud integration can allow for historical data tracking, trend analysis, and automated reporting. Coupled with machine learning algorithms, the system could even predict pollution trends and identify potential sources of contamination.

10.Black Box Testing and White Box Testing

Black Box Testing

Black Box Testing involves system testing and does not analyze its internal code, circuit connections or hardware design. During this type of testing, the internal working of the system is not taken into consideration. Rather, the system is fed with inputs and the outputs monitored to determine whether the system is behaving as desired in regard to the stipulated requirements.

Black Box Testing was applied in this project to check the functionality of RC Water Quality Monitoring Boat to give the appropriate movement commands and place it in the different water situations. To be tested, the system is made up of the Arduino Uno, HC-05 Bluetooth, turbidity sensor, DC motors with motor driver, and the mobile phone application. In the testing, the tester does not open the Arduino program or look at the wiring connections. The tester only interacts with the system via the mobile application whereby commands are sent and the movement of the boat and the values of turbidity are shown on the screen.

A number of test scenarios were carried out to test the effectiveness of the system. The HC-05 Bluetooth module was coupled with the mobile phone in Bluetooth Connectivity Test. The anticipated result was to be able to make a successful Bluetooth connection between the mobile device and the system. The result obtained proved that the connection was made successfully and it was possible to transmit commands to the Arduino with no errors.

In Forward Movement Test, Forward command was transmitted through mobile application. It was anticipated that the two motors would be rotating forward making the boat move forward. The results observed were that the boat went forward in a smooth movement as it was supposed to.

Backward command worked on the mobile application in the Backward Movement Test. The anticipated result was that the motors would turn backwards causing the boat to move backwards. The outcome realized was the correct movement of the boat in the reverse direction.

Stop command was given in the Stop Command Test. The anticipated deliverables were the immediate cessation of the motors. The experimented output revealed that the boat had come to a complete halt.

The sensor was tested using the Turbidity Sensor Test whereby the boat was first put in clean water and then in a muddy one to determine the behavior of the sensor. It was anticipated that the turbidity values in clean water were low and muddy water contained high values of turbidity. The results obtained indicated that the sensor reading was able to vary depending on the water condition, and the values were also shown properly on the mobile screen.

The Arduino was connected to the mobile phone via the Bluetooth module to transmit sensor data to the phone in the Communication Test. The anticipated deliverable was to have real-time show of the turbidity values on the mobile application. The result obtained proved that the mobile application was able to receive and present the turbidity readings.

All in all, the Black Box Testing procedure proved that the RC Water Quality Monitoring Boat can react appropriately to user commands and measure and send proper data on the water quality without considering the way of how the system works internally.

White Box Testing

White Box Testing was conducted to test the internal functionality of both hardware and software used to make the Water Pollution Monitoring RC Boat system. This testing was mainly aimed at making sure that the Arduino program, Bluetooth communication, processing of the turbidity sensor as well as the motor control logic was functioning properly and in the right coordination with the rest and also to ensure that these specific logical operations were working properly together.

The Arduino Uno program was first checked which ensured that all inputs and output pins were defined in the program. The pins associated with the turbidity sensor, Bluetooth module, and motor driver were thoroughly verified in the program and matched with the real physical connections to be applied on the breadboard. This measure has made sure that sensor and motor control signals are correctly mapped and connected to the right parts.

The Arduino program setup section was then questioned to ensure that there was proper initiation of Bluetooth serial communication. It was also confirmed that motor driver control pins were set with the output pins. This guaranteed that the system will be ready to respond to commands and turn the motors on right after the supply of power.

Then, the logic of processing the Bluetooth commands was examined. The internal program flow was followed through the sending of the movement commands with the mobile application to ensure that the received characters were properly stored and matched with the predefined conditions of the code. This ensured that movement of the boat is correct in respect to a command as given by a user.

An internal test was also done on the turbidity sensor reading section of the code. The analogy value feeds of the sensor were followed to make sure that they varied accordingly as the sensor was used in muddy and clean water. The program code that manipulates these values in calculating the level of turbidity was checked to make sure that the values are interpreted correctly and the data is sent via Bluetooth.

Also the motor control logic was discussed. The signal flow path between the output pins of the Arduino and the motor driver input was tested to ensure that it was operating correctly. It was also established that once a certain movement command is given, the respective signals were sent to the two motors enabling them to move in the desired direction and the boat moved smoothly.

Lastly, the program section on data transmission was also checked to make sure that the values of turbidity were continuously transmitted to the mobile phone through Bluetooth without failure, loss or delay of data.

In this comprehensive internal testing of the program flow, wiring connections, and control logic, the White Box Testing established that the internal parts of the RC Boat system have been correctly integrated and that they work reliably to provide real time water pollution monitoring.

11.TEST CASES, TEST COVERAGE AND TESTING TOOLS

TEST CASES

Testable Unit	Test Case Description	Value Entered (Example)	Expected Output	Result
Bluetooth Module	Connect mobile to HC-05	Bluetooth ON, Pair HC-05	Bluetooth connected successfully	Pass
Motor Control	Forward movement command	Command = 'F'	Boat moves forward	Pass
Motor Control	Backward movement command	Command = 'B'	Boat moves backward	Pass
Motor Control	Stop command	Command = 'S'	Boat stops	Pass
Turbidity Sensor	Test in clean water	Clear water	Low turbidity value displayed	Pass
Turbidity Sensor	Test in muddy water	Muddy water	High turbidity value displayed	Pass
Data Transmission	Send sensor value	Turbidity = 320	Value shown on mobile	Pass
Power Supply	Power ON system	9V battery	System starts normally	Pass
Motor Driver	Both motors ON	Forward command	Both fans rotate	Pass
Bluetooth Communication	Continuous data flow	Repeated sensor values	Data received without loss	Pass

Test Coverage

Test coverage is a concept which shows how much a system has been tested in the course of testing. It assists in knowing whether or not all the key elements and functionalities of the system have been tested.

In the case of the RC Boat Water Pollution Monitoring System, the major hardware and software modules were incorporated in the test procedure to ascertain the correct system validation. The Bluetooth communication module was tested in order to confirm that it was pairing, transmitting the commands, and receiving the data between the mobile device and the Arduino system. Motor control system was also tested to make sure that there are forward, backward movements, and stop movements according to the user commands.

To ascertain correct sensor reading and variation in sensor readings, the turbidity sensor was experimented with clean and polluted water samples. The power supply system was also looked into to have steady power supply to both the Arduino board and the motor driver. Furthermore, the functionality of data display was also checked in terms of whether the turbidity values have been properly displayed on the mobile application in real time.

The system passed high test coverage in terms of functionality and performance as all key modules of the RC boat were tested such as communication, sensing, control, and power management.

Testing Tools Used

Different tools and practical materials were employed in the experiment of the system in order to correct validation of the hardware and software components.

Arduino program code was also written, uploaded and verified by the Arduino IDE. Serial Monitor on the Arduino IDE was useful in monitoring Bluetooth communication and sensor data values in internal testing. The control was done by sending movement commands to the boat and receiving readings of the turbidity sensor in real time on an android device via a Bluetooth Terminal application.

The sample of the turbidity sensor was tested on the performance and accuracy under varying conditions of environmental clean and muddy water. Lastly, the prototype of the RC boat was tested in real-time based on the control of their movements, the transmission of sensor data, and the general systems integration.

All of these tools were used to test the entire Water Pollution Monitoring RC Boat system effectively and reliably.

12. VERIFICATION AND VALIDATION PROCESS

Verification

Verification refers to checking whether the system has been developed properly in accordance with approved system specifications and design. It also aims at ensuring the correct implementation of all the hardware and wiring connections and programming logic of the RC boat according to plan. Verification on each development stage was performed in this project so that the RC boat, the turbidity sensor, the Bluetooth communication module, the Arduino controller, and the motor system were assembled and integrated accordingly.

1. Verification Process

The verification process was initiated by verifying all the hardware components to the system design diagram. The Arduino Uno, turbidity sensor, HC-05 Bluetooth module, motor driver, DC motors, and batteries were checked in order to confirm that the right components are chosen and applied according to the demands of the project.

The wiring connections were then checked upon with a thorough scrutiny after verification of the components. Every physical connection was matched with the circuit diagram so as to be correct. Power supply lines, ground connection, motor driver input pins, Bluetooth TX and RX connection and sensor input pin were checked to avoid wrong wiring and possible faults.

Arduino program was then studied in detail. The pin configurations, Bluetooth command processing logic, the turbidity sensor reading section, and the motor control instructions were assessed to make sure that the software implementation was readable to the system design. This was done to ensure that the internal logic of the program matched with the functional requirements of the project.

2. Reviews

The system had regular reviews that were made after they had completed a certain module. The project team and the project guide really considered the boat structure, the position of its motor, the arrangement of wiring, the process of Bluetooth pairing, and the position of the sensors. These reviews have assisted in the detection of small problems at an early stage before bigger problems are realized in the final testing.

3. Audit

Internal audit was conducted prior to final testing to check in general system readiness. All the hardware connections, program logic and power supply setups were carefully verified. The boat was started and all modules were monitored to be sure that there was no overheating, short circuit or any abnormal behavior. This auditing also made the system safe and stable to operate in real time.

4.Meetings with Project Guide

The project guide would meet with them on a regular basis to talk about the progress of development and technical difficulties. Problems with motor performance, failures in connecting to the Bluetooth technology as well as irregular sensor measurements were addressed in detail. Similar to other meetings, corrective measures and improvements were made based on the recommendations given in such meetings in order to improve system performance and reliability.

5.Walkthrough

The operation of the entire system was also systematically inspected by running the entire system through steps. This involved setting the Bluetooth connection, transmitting movement commands on the mobile application, monitoring motor response, monitoring turbidity sensor readings, and data transmission to the mobile device. This sequential assessment made sure that all the stages of the system worked as per the workflow and functional plan.

Peer Review Report

A peer review was done to analyze the general design, functionality, and reliability of this project, which is Water Pollution Monitoring Using RC Boat. This review was done to evaluate the hardware configuration, performance of the sensor, system of communication and the general integration of the system.

As part of the review, the RC boat framework was discovered to be well-balanced and able to bear all electronic parts without loss of balance. The location of the Arduino Uno, turbidity sensor, Bluetooth module, motor driver, and batteries were appropriate and allowed the smooth and safe working. The wiring was well laid and clean and this reduced chances of loose wires and short circuiting.

The turbidity sensor was tested regarding its accuracy and correct location in water. As it was observed, the sensor was able to sense the differences between clean and muddy water. The Bluetooth module was able to communicate hardware and software components effectively because the sensor readings were correctly sent to the mobile phone.

Motor, propulsion system was also checked. Both motors reacted well to movement instructions allowing forward, backward and stop movement. Once minor problems with wiring were solved, motors were turning properly in sync with each other and the boat could be maneuvered freely.

The Arduino code was reviewed to make sure that it properly dealt with Bluetooth control messages, sensor readings, and motor control code. The organization of the program proved to be understandable, organized, and effective in terms of the needed operations.

On the whole, the peer review found that the project will achieve its goal of offering a safe and low-cost, and reliable solution to monitor water pollution. The system was considered to be workable, well-integrated and appropriate in real-time water quality monitoring application.

Validation

The process that can be used to verify that the developed system meets user requirements and operates effectively in the real-world operating conditions is called validation. Whereas the verification aspect concentrates on the correct construction of the system, validation is concerned with whether the system is being constructed in the right way. In this project, validation was conducted to ensure that the RC Boat is a practical and meaningful solution towards ensuring that water pollution is monitored.

The process of validation started by taking the RC boat to a real-life situation in the real water and running it with the help of a mobile phone through Bluetooth. There were forward, backward and stop commands in the boat which were used to control it. The boat was noted to react well and easily to every command given to it and this showed that the system is easily operated by a user who is not required to physically interact with water. This ascertained that the remote-control functionality was effective.

The validity of the turbidity sensor was then tested by assessing the boat at varying water samples. The turbidity values on the mobile phone were low when the boat was put in clean water. Conversely, turbidity values were recorded to be high in conditions where the boat was put in muddy water or polluted water. This was a clear indication that the sensor can be used to identify changes to the quality of water and precisely record the level of pollution. These were verified to indicate that the system is undertaking meaningful environmental monitoring.

The wireless communication between the mobile device and the Arduino was also confirmed. The Bluetooth module was able to transmit real time turbidity data without any noticeable delay or loss of data. This made the system reliable in providing uninterrupted live surveillance and observation by the user.

In addition, the entire system was put into trials of practical usage conditions which comprised of battery-powered operation, flow of water, and transmission of data which is continuous. The RC boat was stable in performance during the testing process. No critical operational problems were experienced, and all aspects were cooperating well.

In general, the process of validation proved that the Water Pollution Monitoring RC Boat fits user needs and accomplishes its goal of the provision of safe, remote, cost-effective, and efficient solution to water pollution monitoring in the real environment.

13. Decision Table and Boundary Value Analysis

Decision Table

Module: Arduino Control Logic

1. Purpose

The decision table is prepared to represent the logical behavior of the system based on different input conditions. It helps in understanding how the RC boat responds to user commands and water quality conditions.

2. Identified Conditions

For the hardware system, two primary conditions were considered:

- **C1:** Movement Command Given (Yes / No)
- **C2:** Turbidity High (Yes / No)

3. Identified Actions

Based on the above conditions, the system performs the following actions:

- **A1:** Move Boat
- **A2:** Stop Boat
- **A3:** Display “Polluted Water”
- **A4:** Display “Clean Water”

4. Decision Table ($2^2 = 4$ Rules)

Since the system has 2 conditions, total possible combinations: 4 Rules

Conditions / Rules	R1	R2	R3	R4
C1: Movement Command	Yes	Yes	No	No
C2: Turbidity High	Yes	No	Yes	No
Actions				
A1: Move Boat	✓	✓	—	—
A2: Stop Boat	—	—	✓	✓
A3: Display Polluted	✓	—	✓	—
A4: Display Clean	—	✓	—	✓

5. Interpretation

- If movement command is given and turbidity is high, the boat moves and polluted water is displayed.
- If movement command is given and turbidity is low, the boat moves and clean water is displayed.
- If no movement command is given and turbidity is high, the boat stops and polluted water is displayed.
- If no movement command is given and turbidity is low, the boat stops and clean water is displayed.

6. Conclusion

The decision table clearly defines system behavior under all possible combinations of input conditions. It ensures proper control of boat movement and accurate water quality indication.

Boundary Value Analysis

1.Introduction

Boundary Value Analysis is the testing methodology applied to test the behavior of the system with extreme or critical values of inputs. This method does not test normal values only and instead, the limits during which a more probable error is likely to occur are considered.

2.Application in Our Project

Boundary Value Analysis was used to the turbidity sensor values in our project, Water Pollution Monitoring Using RC Boat.

The turbidity sensor is used to determine the clarity of water and depending on a set threshold value, the water is either:

- Clean Water
- Polluted Water

The fact that there is a change in the classification at a set threshold implies that one should test values at or close to the boundary.

3.Boundary Conditions Being Referred to

The following values were tested to check the appropriate system behavior:

- A score that is slightly below the threshold.
- The exact threshold value

- A figure above the target marginally.

The values assist in making sure that the system does not give wrong or unstable values when the water condition is altered around the decision point.

4.Observations

During testing:

- At the cases when the turbidity was lower than the threshold, the system showed clean water properly.
- As soon as the value went past the threshold, the system dealt with the transition appropriately.
- In cases where the value was greater than the threshold, polluted water was shown.
- No unexpected switching or wrong readings were recorded around the edge.

5.Conclusion

Boundary Value Analysis supported the fact that system can detect the quality of water correctly at critical limit values and can be used reliably in edge cases.

14.Data Flow Testing and Path Testing

DATA FLOW TESTING

Module Used: Arduino Control & Water Quality Monitoring Module

Pseudo Code

1. Start
2. Read turbidityValue
3. If turbidityValue > threshold
4. waterStatus = "Polluted"
5. Else
6. waterStatus = "Clean"
7. Read movementCommand
8. If movementCommand == ON
9. Move Boat
10. Else
11. Stop Boat
12. Display waterStatus
13. End

Definition (d), Use (u), Kill (k)

Variable	d (Definition)	u (Use)	k (Kill)
turbidityValue	Step 2	Step 3	New reading
waterStatus	Step 4, 6	Step 12	Recalculated
movementCommand	Step 7	Step 8	New command

Data Flow Paths

Path 1 – Polluted Water + Movement ON

1 → 2 → 3 → 4 → 7 → 8 → 9 → 12 → End

- d(turbidityValue)
- u(turbidityValue)
- d(waterStatus)
- u(waterStatus)
- d(movementCommand)
- u(movementCommand)

Path 2 – Clean Water + Movement ON

1 → 2 → 3 → 6 → 7 → 8 → 9 → 12 → End

- d(turbidityValue)
- u(turbidityValue)
- d(waterStatus)
- u(waterStatus)
- d(movementCommand)
- u(movementCommand)

Path 3 – Movement OFF

1 → 2 → 3 → (4 or 6) → 7 → 8 → 11 → 12 → End

- d(turbidityValue)
- u(turbidityValue)
- d(waterStatus)
- u(waterStatus)
- d(movementCommand)
- u(movementCommand)

PATH TESTING

Pseudo Code

1. Start
2. Check Bluetooth connection
3. If Bluetooth not connected
4. Stop boat
5. Else
6. Read movement command
7. If command == Forward
8. Move boat forward
9. Else if command == Backward
10. Move boat backward
11. Else
12. Stop boat
13. Read turbidity sensor
14. Send turbidity data to mobile
15. End

Control Flow Graph (Nodes)

Node	Statement
1	Start
2	Check Bluetooth
3	Bluetooth decision
4	Stop boat
6	Read command
7	Forward decision
8	Move forward
9	Backward decision
10	Move backward
12	Stop boat
13	Read turbidity
14	Send data
15	End

Edges

1 → 2
2 → 3
3 → 4
3 → 6
4 → 15
6 → 7
7 → 8
7 → 9
9 → 10
9 → 12
8 → 13
10 → 13
12 → 13
13 → 14
14 → 15

Cyclomatic Complexity

Formula:

$$V(G) = E - N + 2$$

$$E = 15$$

$$N = 13$$

$$V(G) = 15 - 13 + 2$$

$$V(G) = 4$$

OR

$$V(G) = \text{Number of Decision Nodes} + 1$$

Decision Nodes:

- Bluetooth connected?
- Forward command?
- Backward command?

So,

$$V(G) = 3 + 1 = 4$$

Independent Paths

Since complexity = 4

We need **4 independent paths**.

Path 1 – Bluetooth Not Connected

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 15$$

Condition: Bluetooth not connected

Path 2 – Forward Movement

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 6 \rightarrow 7 \rightarrow 8 \rightarrow 13 \rightarrow 14 \rightarrow 15$$

Condition:

- Bluetooth connected
- Command = Forward

Path 3 – Backward Movement

1 → 2 → 3 → 6 → 7 → 9 → 10 → 13 → 14 → 15

Condition:

- Bluetooth connected
- Command = Backward

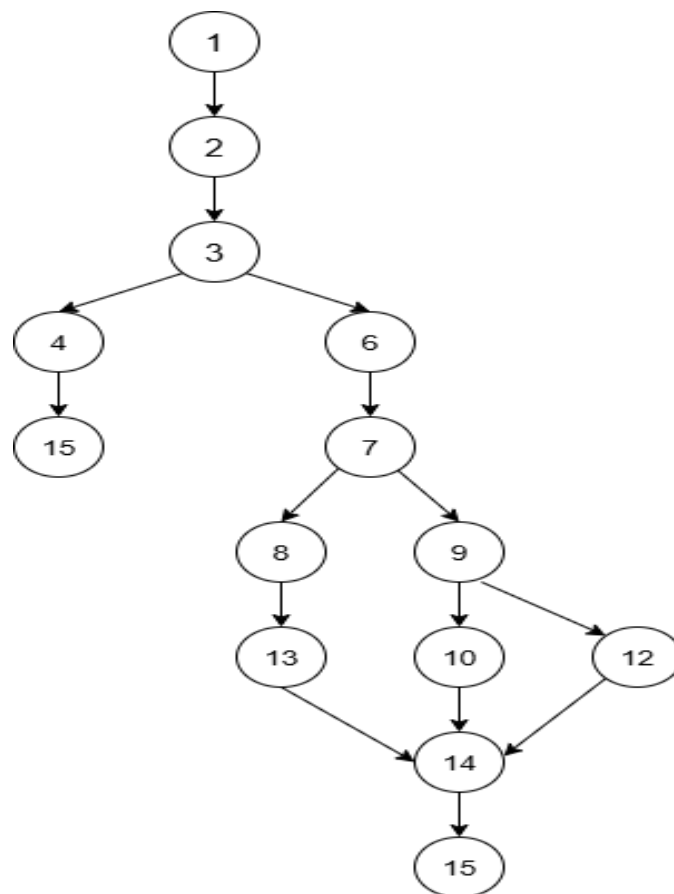
Path 4 – No Movement Command

1 → 2 → 3 → 6 → 7 → 9 → 12 → 13 → 14 → 15

Condition:

- Bluetooth connected
- No valid movement command

Control flow Diagram



15.Online Testing Tool Used

Tool Name: Serial Bluetooth Terminal Application

Purpose of Tool

The wireless communication between the mobile device and the Arduino-based hardware system was tested with the help of the Serial Bluetooth Terminal application.

It helped in:

- Transfer of movement instructions to the RC boat.
- Getting the reading of the turbidity sensor.
- Checking the transmission of real-time data.
- Ensuring the stability of Bluetooth connectivity.

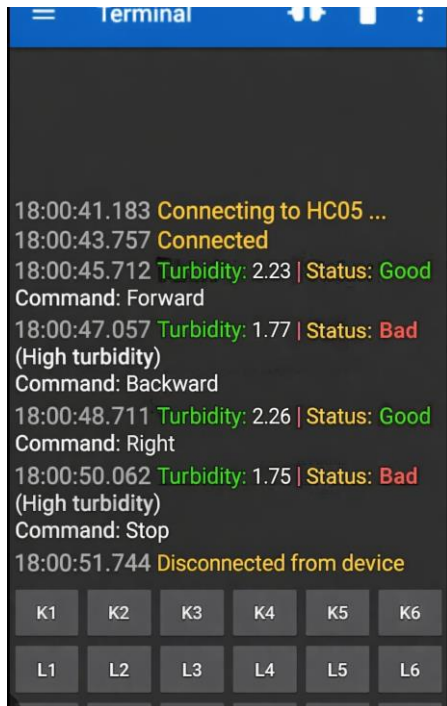
Testing Procedure

1. The HC-05 Bluetooth module was turned on and connected with the mobile device.
2. Serial Bluetooth Terminal application was started.
3. A relationship was also built with the HC-05 module.
4. Such commands as Forward, Backward and Stop were relayed.
5. The terminal screen saw the values of turbidity sensor.
6. Tests on several commands were undertaken to confirm the reliability and response time.

Observations

- The boat reacted appropriately to movement instructions.
- The values of turbidity sensors were shown in real-time.
- There was no major delay in the execution of the commands.
- Communication was controlled at normal operating range.

Screenshot Description



In the screenshot, the Bluetooth connection with the HC-05 module is successful and displays the values of the turbidity sensors in real time.

Conclusion

Serial Bluetooth Terminal application was an efficient testing tool in the hardware based project to test wireless communication, command execution and sensor data transmission.

16. References:

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HC-05 Bluetooth Module Datasheet. Available online: <https://www.electronicwings.com>

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